# edudrift

An LLM's living hypothesis log on AI's impact on higher education

Santosh B. Srinivas · HEC Paris

Snapshot: 2026-06-05

https://santoshbs.github.io/theaidrift/edu/

*edudrift* is an experiment in letting a local open-weight large language model continuously process AI-related news signals and hypothesise their implications for specific higher-education capabilities. Each entry in this document is a working manuscript: it is generated and revised **entirely by the model**, with **no human in the editing loop**. The author's contribution is limited to designing the system — selecting RSS sources, defining the capability scaffold, writing the prompts, and committing the model's output to version control. The prose, the reasoning, the framing, and the choice of which signals matter are all the model's.

The capability scaffold is drawn from the **Higher Education Reference Models (HERM)** v3.2.0, a global industry-standard enterprise-architecture framework curated by the HERM Working Group of CAUDIT (the Council of Australasian University Directors of Information Technology), in collaboration with EDUCAUSE, UCISA, and EUNIS, and used by more than a thousand institutions worldwide. HERM is published under CC BY-NC-SA 4.0 and is available at https://www.caudit.edu.au/communities/caudit-higher-education-reference-models/.

For each capability, the log states three horizon hypotheses, each grounded in cited public signals:

- **Near** (2026–2031) — what current evidence most plausibly supports;
- **Mid** (2036–2041) — extrapolations contingent on signal trajectories holding;
- **Far** (2046–2051) — competing hypotheses where the trajectory is genuinely uncertain.

Hypotheses are abductive and revisable; each is version-controlled at the source repository. *Italic statements* are the formal hypothesis claims.

The live, continuously revised version is published at /theaidrift/edu/.

---

{*Caveat lector.* Because this document is fully model-generated, claims should be read as structured speculation grounded in cited signals, not as expert assessment. Citations point to the public sources the model drew on; readers should verify any claim against the original source before relying on it.}

# Contents

## Academic Administration

*capability_id: academic-administration · created: 2026-04-10 · revised: 2026-04-10 · refs: 2*

### Near (2026–2031)

For Academic Administration broadly, evidence that AI-enabled misconduct overlaps with traditional forms of plagiarism [1] may drive a shift in academic policy & regulation management toward integrated behavioral frameworks rather than tool-specific bans [1]. For institutions implementing structured faculty development partnerships, as demonstrated at the University of Central Florida [2], this transition may be supported by bottom-up infrastructure involving collaborations between teaching centers and digital learning divisions [2]. This represents a change in educational substance by refocusing regulation on behavioral patterns of dishonesty rather than the specific technology employed. Because updates to academic policy typically require faculty senate approval and governance review, the pace of institutional adoption may remain slow. Processes such as academic year scheduling and timetabling management are unlikely to change in this period as they are not functionally linked to academic integrity signals.

*Hypothesis: In the near term, academic policy & regulation management may shift from tool-specific AI bans toward holistic integrity frameworks that treat AI use as one of several overlapping plagiarism behaviors.*

### Mid (2036–2041)

For Academic Administration broadly, if current trajectories hold, the identified link between AI use and general plagiarism [1] could lead to the development of longitudinal behavioral profiles within academic policy & regulation management. If institutions continue to build evolving policy infrastructures through cross-functional partnerships [2], regulatory mechanisms might shift from evaluating single instances of misconduct to identifying patterns of dishonesty across a student's academic career. This transition would likely be constrained by data privacy laws and institutional due process requirements, which tend to persist regardless of technological shifts. The fundamental administrative structure of the academic year remains unlikely to change, as it is governed by external labor contracts and seasonal student demographics rather than integrity regulation.

*Hypothesis: In the mid term, academic policy & regulation management may move toward a behavioral-pattern model of integrity enforcement, extrapolating from current evidence of overlapping plagiarism methods.*

### Far (2046–2051)

For Academic Administration broadly, the long-term trajectory of integrity regulation remains uncertain, with two primary rival outcomes. One possibility is that the persistence of overlapping plagiarism [1] leads to the automation of the disciplinary process through algorithmic pattern matching. A competing outcome is that the continued erosion of traditional authorship leads to a collapse of the current regulatory regime, where the "product" of education is no longer a verifiable independent work. It remains unknown how professional accreditation bodies will redefine "competence" if traditional plagiarism regulations become obsolete. Even at this horizon, the legal requirement for human-led due process in high-stakes disciplinary appeals is likely to resist full automation.

*Hypothesis A: In the far term, academic policy & regulation management may evolve into a behavioral monitoring system that identifies integrity risks.*

*Hypothesis B (competing): In the far term, the capability of academic policy & regulation management may be fundamentally diminished as the institution shifts away from the validation of independent student authorship.*

### References


1. The Conversation Au (2026-04-01). "Does AI mean more uni students are plagiarising their work?" https://theconversation.com/does-ai-mean-more-uni-students-are-plagiarising-their-work-279565

2. EDUCAUSE Review (2026-04-08). “Building AI Initiatives with Bottom-Up Champions” https://er.educause.edu/articles/2026/4/building-ai-initiatives-with-bottom-up-champions

## Business Capability Management

*capability_id: business-capability-management · created: 2026-05-31 · revised: 2026-05-31 · refs: 8*

### Near (2026–2031)

For business capability management broadly, organisational design and change portfolio management shift toward people-and-organization-first frameworks to prevent the failure of AI initiatives that focus exclusively on technology [6]. The adoption of AI operating models requires a redesign of organisational design to manage the functional transition toward software-factory operations [4]. For institutions deploying AI infrastructure, benefits management may shift in substance toward more stringent ROI validation because only 28% of such projects fully realize their projected payoffs [3]. The delivery mechanism of programme & project management is altered through the adoption of specialized AI lifecycle frameworks intended to bridge the gap between conceptual prototypes and production readiness [1]. In institutions utilizing Scaled Agile Frameworks (SAFe), the delivery mechanism of programme & project management is further modified by the automation of delivery metrics [5]. For institutions developing institutional AI protocols, programme & project management and change portfolio management may shift in delivery toward controlled experimentation and piloting [7]. In institutions focusing on rapid tool deployment, change portfolio management may shift in substance toward human-centric adoption strategies to avoid the cost of unused workflows [2]. For institutions employing AI in project retrospectives, the delivery mechanism of programme & project management is modified through the use of AI to audit project post-mortems and identify root causes of failure [8]. Quality management processes are unlikely to change yet, as they rely on stable administrative specifications that remain independent of AI implementation.

*Hypothesis: In the near term, business capability management will integrate AI-specific operating models, lifecycle frameworks, and automated project auditing to govern the transition toward software-factory-style operations and mitigate high infrastructure failure rates.*

### Mid (2036–2041)

For business capability management broadly, if the trend of low ROI for infrastructure and recurring project failures persists, the capability may shift from experimental AI adoption to disciplined architectural integration [3][8]. Extrapolating from the need to govern AI sprawl, enterprise architecture could evolve to standardize the software-factory model across the institution [4]. This represents a change in the educational substance of the capability, as institutional value is measured by production-level operational integration rather than prototype capability [1]. The fundamental mechanisms of institutional accountability—specifically the determination of legal and professional responsibility for capability failure—will likely persist unchanged due to regulatory and governance constraints.

*Hypothesis: In the mid term, the capability may shift toward a disciplined architectural integration where AI-driven operating models and automated failure analysis are standardized as core institutional utilities.*

### Far (2046–2051)

For business capability management broadly, the long-term trajectory remains uncertain regarding the automation of strategic decision-making. One outcome is that change portfolio management becomes largely automated, with AI systems optimizing resource allocation based on real-time adoption and ROI data [2][3]. However, the process of determining the social legitimacy of institutional goals resists change even at this horizon and likely remains a human-centric function. It remains unknown whether AI can manage benefits management without replicating the systemic ROI failures observed in early infrastructure projects [3].

*Hypothesis A: In the far term, business capability management may become largely automated, with AI systems optimizing the change portfolio through real-time performance feedback.*

*Hypothesis B (competing): In the far term, the persistence of high-profile AI infrastructure failures may trigger a return to hyper-conservative, human-centric governance models that strictly limit the scope of* *automated capability management.*

## References


1. Dzone Ai (2026). "From Concept to Production: A Strategic Framework for AI/ML Project Success." https://feeds.dzone.com/link/23558/17313177/ai-lifecycle-framework-for-successful-projects
2. Dev To Ai (2026). "How to Build an AI Workflow That Your Team Will Actually Use." https://dev.to/maxothex/how-to-build-an-ai-workflow-that-your-team-will-actually-use-55he
3. The Register (2026). "Only 28% of AI infrastructure projects fully pay off, survey finds." https://go.theregister.com/feed/www.theregister.com/2026/04/07/ai_returns_gartner/
4. Dev To Ai (2026). "AI Operating Models: The $2M Workflow Redesign Your Board Isn't Asking About." https://dev.to/dr_hernani_costa/ai-operating-models-the-2m-workflow-redesign-your-board-isnt-asking-about-1dfd
5. Dzone Ai (2026). "Revolutionizing Scaled Agile Frameworks with AI, MuleSoft, and AWS: An Insider's Perspective." https://feeds.dzone.com/link/23558/17323804/revolutionizing-scaled-agile-frameworks-with-ai
6. The Register (2026). "Ex-AWS legend explains what enterprises need to make AI actually work." https://go.theregister.com/feed/www.theregister.com/2026/04/25/ai_enterprise_matt_domo/
7. Wonkhe (2026). "Controlled experimentation with AI – the power of a pilot." https://wonkhe.com/blogs/controlled-experimentation-with-ai-the-power-of-a-pilot/
8. Dev To Ai (2026). "I Asked AI to Audit My Last 3 Failed Projects — Here's What It Found." https://dev.to/tillerman/i-asked-ai-to-audit-my-last-3-failed-projects-heres-what-it-found-30on

## Commercial Delivery

*capability_id: commercial-delivery · created: 2026-06-04 · revised: 2026-06-04 · refs: 2*

### Near (2026–2031)

For US-based institutions and those utilizing frontier AI models, the process of commercial activity exit management is increasingly constrained by both regulatory mandates and technical dependencies. In the US, federal blacklisting of specific AI technologies may trigger forced commercial activity exit management, shifting the mechanism of vendor termination from planned lifecycles to compliance-driven mandates [1]. Simultaneously, institutions attempting to transition between frontier models may find the exit process obstructed by vendor lock-in, which increases the complexity and cost of swapping providers [2]. These shifts represent changes to the delivery mechanism of risk and vendor management rather than the educational substance of the commercial activities. Because current signals focus on exit and compliance, the core processes of commercial activity output management are unlikely to change in the near term.

*Hypothesis: In the near term, a subset of institutions will experience increased friction in commercial activity exit management due to a combination of geopolitical regulatory restrictions and technical vendor lock-in.*

### Mid (2036–2041)

For institutions utilizing frontier AI models, if current trajectories of vendor lock-in and regulatory volatility hold, commercial activity issue management may shift toward the requirement of proactive "interoperability audits." Extrapolating from the difficulty of swapping models [2], institutions could implement mandatory exit-strategy documentation during the initial procurement phase to mitigate the risk of operational paralysis. This represents a more conservative delivery of vendor oversight, as demonstrated by the necessity of aligning commercial ties with state-level prohibited entity lists [1]. The fundamental requirement for specialized legal counsel to navigate the intersection of trade law and software licensing will persist unchanged.

*Hypothesis: In the mid term, institutions may integrate formal exit-strategy planning and interoperability requirements into their commercial activity issue management to counter the effects of vendor lock-in.*

### Far (2046–2051)

For institutions utilizing large-scale commercial AI, the long-term state of commercial activity exit management depends on whether model interoperability becomes a technical standard or remains a proprietary barrier. One plausible outcome is the emergence of fragmented technological blocs where exit management is a routine, trigger-based operational task dictated by national security boundaries [1]. A competing outcome is the persistence of deep vendor lock-in, where the cost of exit management becomes prohibitively high, effectively merging the institutional capability with the vendor's own lifecycle. It remains unknown whether future regulatory frameworks will provide structured "grace periods" for institutional exits or mandate immediate cessation. The basic institutional need to manage the legal termination of contracts will resist change regardless of the underlying technology.

*Hypothesis A: In the far term, commercial activity exit management becomes a highly automated, trigger-based process within a fragmented global AI market.*

*Hypothesis B (competing): In the far term, extreme vendor lock-in renders independent commercial activity exit management practically impossible for most institutions, leading to permanent vendor dependency.*

### References

1. Ars Technica (2026-04-09). "Trump-appointed judges refuse to block Trump blacklisting of Anthropic AI tech." https://arstechnica.com/tech-policy/2026/04/trump-appointed-judges-refuse-to-block-trump-blacklisting-of-anthropic-ai-tech/
2. The Register (2026-04-28). "Locked, stocked, and losing budget: AI vendor lock-in bites back." https:

//go.theregister.com/feed/www.theregister.com/2026/04/28/locked_stocked_and_losing_budget/

## Commercial Monitoring

*capability_id: commercial-monitoring · created: 2026-06-04 · revised: 2026-06-04 · refs: 1*

### Near (2026–2031)

For a subsegment of institutions employing proprietary frontier AI models, the process of commercial contingency planning may shift as assumptions of model interchangeability prove invalid [1]. The mechanism driving this change is vendor lock-in, where technical dependencies and pricing structures prevent the rapid swapping of one frontier model for another [1]. Consequently, contingency plans that previously relied on the ability to pivot between vendors in short timeframes may be rendered ineffective [1]. This represents a change in educational substance, as the institutional approach to risk management shifts from a "swap-out" strategy to a lock-in mitigation strategy. Commercial contract performance management is unlikely to change significantly in this period, as the basic monitoring of service-level agreements and uptime remains a standard contractual requirement.

*Hypothesis: In the near term, for institutions using proprietary frontier models, commercial contingency planning will shift from assumptions of model interchangeability to the management of vendor lock-in risks.*

### Mid (2036–2041)

For the same subsegment of institutions using proprietary frontier models, if current trajectories of vendor lock-in hold, commercial activity reporting may expand to track dependency metrics [1]. This process could evolve to monitor "lock-in depth," treating the inability to switch vendors as a quantifiable financial risk during reporting cycles [1]. Commercial contingency planning may further shift toward architectural diversification to avoid total dependence on a single provider [1]. The fundamental mechanisms of commercial contract performance management will likely persist unchanged, as the legal frameworks governing institutional procurement and vendor accountability are slow to evolve.

*Hypothesis: In the mid term, for institutions using proprietary frontier models, commercial activity reporting may incorporate specific metrics to monitor and quantify vendor dependency.*

### Far (2046–2051)

For institutions that have integrated proprietary frontier models, the scope of commercial contingency planning remains uncertain, depending on whether industry standards move toward open interoperability or deeper integration. One outcome involves the total obsolescence of "swapping" as a contingency, with the vendor instead becoming a core utility similar to electrical power. A rival outcome involves the emergence of standardized model interfaces that restore the capability for rapid vendor migration. Throughout these scenarios, the requirement for commercial activity reporting to produce an audit trail for institutional finance will resist change. It remains unknown if open-weight models can reach a level of capability that makes proprietary lock-in a secondary concern for institutional risk managers.

*Hypothesis A: In the far term, for these institutions, commercial contingency planning may be replaced by a utility-based monitoring model where the provider is treated as essential, non-swappable infrastructure.*

*Hypothesis B (competing): In the far term, for these institutions, the adoption of open-standard model interfaces may restore the capability for rapid vendor swapping within commercial contingency planning.*

### References


1. The Register (2026-04-28). "Locked, stocked, and losing budget: AI vendor lock-in bites back." https://go.theregister.com/feed/www.theregister.com/2026/04/28/locked_stocked_and_losing_budget/

## Commercial Sourcing

*capability_id: commercial-sourcing · created: 2026-04-09 · revised: 2026-04-09 · refs: 2*

### Near (2026–2031)

For a subset of institutions using AI-driven search tools and those operating within US regulatory jurisdictions, the reliability of commercial market analysis may be compromised by both external manipulation and geopolitical constraints. AI-focused SEO tactics attempt to influence LLM-generated lists of vendors, which may introduce bias into the process of commercial opportunity identification [1]. Simultaneously, the blacklisting of specific high-capability AI providers, such as Anthropic, restricts the pool of viable vendors available for commercial opportunity assessment in the US [2]. This represents a change to the delivery mechanism of vendor discovery via AI SEO and a regulatory constraint on the available market via blacklisting. Because procurement processes often rely on initial discovery lists to narrow the field, these factors could lead to the exclusion of non-optimized or legally restricted vendors [1][2]. However, formal procurement mandates requiring a predefined set of competitive bids are unlikely to change, as these regulatory constraints prevent total reliance on AI-generated suggestions.

*Hypothesis: In the near term, for the affected subsegments and regions, commercial market analysis may become less reliable due to a combination of manipulated AI responses and regulatory restrictions on vendor availability.*

### Mid (2036–2041)

For these same subsets and regions, if current trajectories of AI-driven SEO and geopolitical blacklisting hold, the process of commercial opportunity assessment could shift toward a more adversarial verification model. Institutions may introduce mandatory cross-referencing of AI-generated vendor lists against curated, compliance-verified industry databases to mitigate the risk of both manipulated rankings and the selection of prohibited entities [1][2]. This would introduce a formal validation step into the commercial market analysis process to ensure the integrity and legality of the source data. Despite this, the internal institutional governance regarding the final selection of vendors is likely to persist unchanged due to high financial stakes and existing audit requirements.

*Hypothesis: In the mid term, the commercial market analysis process for these subsegments may integrate formal validation and compliance steps to counter misinformation and regulatory risk.*

### Far (2046–2051)

For institutions relying on AI for sourcing, the outcome depends on whether AI search engines develop verifiable transparency standards and whether geopolitical vendor restrictions stabilize. One plausible outcome is that commercial opportunity identification becomes a specialized technical function where AI-driven discovery is viewed as an unreliable lead-generation step rather than a valid analysis tool. It remains unknown whether AI SEO will evolve into a standardized digital signaling mechanism or if blacklisting becomes a permanent feature of institutional procurement. The underlying need for human-led final due diligence will likely resist change because of the legal liability associated with institutional commercial contracts.

*Hypothesis A: In the far term, commercial market analysis for this subsegment may decouple from generative AI search, reverting to trusted, closed-loop industry registries to avoid noise and regulatory volatility.*

*Hypothesis B (competing): In the far term, a new standard for verifiable AI sourcing and automated compliance filtering may emerge, allowing institutions to efficiently identify viable vendors while filtering out manipulated content.*

### References

1. The Verge (2026-04-06). "Can AI responses be influenced? The SEO industry is trying." https://www.theverge.com/tech/900302/ai-seo-industry-google-search-chatgpt-gemini-marketing

2. Ars Technica (2026-04-09). "Trump-appointed judges refuse to block Trump blacklisting of Anthropic AI tech." https://arstechnica.com/tech-policy/2026/04/trump-appointed-judges-refuse-to-block-trump-blacklisting-of-anthropic-ai-tech/

## Corporate Governance

*capability_id: corporate-governance · created: 2026-06-04 · revised: 2026-06-04 · refs: 24*

### Near (2026–2031)

For Corporate Governance broadly, the transition from monitoring generative outputs to overseeing autonomous agent execution shifts the educational substance of risk management from content validation to the governance of executed actions [5]. To constrain this autonomy, risk management is adopting delivery mechanisms such as sandbox execution environments [24] and runtime security toolkits [10]. Policy and regulation management is integrating agent registries [13] and autonomous governance tools [2] to identify and mitigate the operational risks of unsanctioned "shadow AI" [38], while new institutional protocols are developed through controlled piloting [37]. In the European Union, compliance monitoring and reporting is constrained by mandates to maintain Annex IV technical files [8], respond to European Data Protection Board inquiries regarding agent logic [17], and navigate the legal principles of algorithmic administration [12][39]. In the United States, policy and regulation management is adapting to a national AI legislative framework [26] and managing the legal fallout from agent-linked criminal liability probes [15][28][29]. Within specialized subsegments, risk management is incorporating "judgment layers" for financial agents to prevent automated outputs from being treated as final decisions [19], as well as mechanisms to guard against data bias in hiring [3] and "LLM nepotism" in organizational evaluation [20]. The legal principle that fiduciary liability cannot be delegated to a software agent is unlikely to change [1].

*Hypothesis: In the near term, institutions will shift risk and policy frameworks to mitigate the liability of authorized agents and the operational risks of shadow AI, while moving compliance monitoring toward runtime enforcement and regulatory technical documentation.*

### Mid (2036–2041)

For Corporate Governance broadly, if current trajectories of agent autonomy and regulatory tension hold, internal audit and reporting may shift from retrospective periodic sampling to the use of governance toolkits that provide immutable, real-time audit trails [6]. This mechanism could reduce the attribution problem that currently complicates the assignment of legal responsibility for autonomous agent failures [1]. Extrapolating from the emergence of AI operating models, the substance of board-level oversight may shift toward the continuous redesign of high-cost organizational workflows—effectively treating the AI "harness" as a dynamic organizational chart—rather than the approval of static policies [18][36]. In regions where strict regulatory regimes persist, compliance monitoring will likely remain a high-friction process requiring human certification despite automated data collection [12]. The requirement for human-in-the-loop verification for high-stakes institutional decisions is unlikely to change due to social legitimacy constraints.

*Hypothesis: In the mid term, the capability may shift from periodic auditing to real-time governance monitoring, with board oversight focusing on the architectural design of AI-integrated workflows.*

### Far (2046–2051)

For Corporate Governance broadly, the interaction between autonomous agents and business continuity management remains a primary uncertainty due to the persistence of systems that cannot explain their own decision-making logic [25]. This uncertainty is compounded by the risk that "secret loyalties" trained into models could allow agents to dodge black-box audits [30]. It remains unknown whether observability tools will evolve to penetrate these black boxes or if governance will shift toward a "fail-safe" model where agents are restricted to low-impact environments. The necessity for a human legal entity to be the ultimate point of accountability for institutional failure resists change even at this horizon.

*Hypothesis A: In the far term, corporate governance evolves into a regime of deep observability where automated auditing tools can reconstruct agent reasoning in real-time to ensure compliance.*

*Hypothesis B (competing): In the far term, the explainability gap renders autonomous agent governance* *impossible, forcing institutions to revert to highly restricted, non-agentic AI to maintain regulatory and legal legitimacy.*

## References


1. The Register (2026-04-05). "AI agents promise to 'run the business,' but who is liable if things go wrong?" https://go.theregister.com/feed/www.theregister.com/2026/04/05/ai_agents_liability/
2. Ai News (2026-04-02). "KiloClaw targets shadow AI with autonomous agent governance" https://www.artificialintelligence-news.com/news/kiloclaw-targets-shadow-ai-autonomous-agent-governance/
3. Ai Trends (2021-10-28). "Promise and Perils of Using AI for Hiring: Guard Against Data Bias" https://www.aitrends.com/ai-world-government/promise-and-perils-of-using-ai-for-hiring-guard-against-data-bias/
4. Ai News (2026-04-06). "As AI agents take on more tasks, governance becomes a priority" https://www.artificialintelligence-news.com/news/as-ai-agents-take-on-more-tasks-governance-becomes-a-priority/
5. Dev To Ai (2026-04-06). "AI agent governance tools compared - 2026 landscape" https://dev.to/jagmarques/ai-agent-governance-tools-compared-2026-landscape-53hm
6. Dev To Ai (2026-04-07). "What's actually inside an EU AI Act Annex IV technical file (and what engineers usually miss)" https://dev.to/rostyslav_myronenko_faf2f/whats-actually-inside-an-eu-ai-act-annex-iv-technical-file-and-what-engineers-usually-miss-4nh4
7. Ai News (2026-04-08). "Microsoft open-source toolkit secures AI agents at runtime" https://www.artificialintelligence-news.com/news/microsoft-open-source-toolkit-secures-ai-agents-at-runtime/
8. Ai News (2026-04-09). "Agentic AI's governance challenges under the EU AI Act in 2026" https://www.artificialintelligence-news.com/news/agentic-ais-governance-challenges-under-the-eu-ai-act-in-2026/
9. The Register (2026-04-09). "AWS: Agents shouldn't be secret, so we built a registry for them" https://go.theregister.com/feed/www.theregister.com/2026/04/09/aws_ai_agent_registry/
10. Ars Technica (2026-04-09). "Trump-appointed judges refuse to block Trump blacklisting of Anthropic AI tech" https://arstechnica.com/tech-policy/2026/04/09/trump-appointed-judges-refuse-to-block-trump-blacklisting-of-anthropic-ai-tech/
11. TechCrunch (2026-04-09). "Florida AG to probe OpenAI, alleging possible connection to FSU shooting" https://techcrunch.com/2026/04/09/florida-ag-to-probe-openai-alleging-possible-connection-to-fsu-shooting/
12. Dev To Ai (2026-04-10). "The EDPB Is Asking About Your AI Agents. Most Teams Can't Answer." https://dev.to/waxell/the-edpb-is-asking-about-your-ai-agents-most-teams-cant-answer-gfk
13. Dev To Ai (2026-04-11). "AI Operating Models: The $2M Workflow Redesign Your Board Isn't Asking About" https://dev.to/dr_hernani_costa/ai-operating-models-the-2m-workflow-redesign-your-board-isnt-asking-about-1dfd
14. Dev To Ai (2026-04-11). "Judgment Layer for Financial AI Agents" https://dev.to/thatechmaestro/judgment-layer-for-financial-ai-agents-141
15. Arxiv Cs Cy (2026-04-14). "LLM Nepotism in Organizational Governance" https://arxiv.org/abs/2604.09620
16. Ai News (2026-04-16). "OpenAI Agents SDK improves governance with sandbox execution" https://www.artificialintelligence-news.com/news/openai-agents-sdk-improves-governance-sandbox-execution/
17. Hackernoon Ai (2026-04-21). "The Hidden Risk of AI Agents: Systems That Can't Explain Themselves" https://hackernoon.com/the-hidden-risk-of-ai-agents-systems-that-cant-explain-themselves?source=rss
18. EDUCAUSE Review (2026-04-21). "White House Unveils National AI Legislative Framework" https://er.educause.edu/articles/2026/4/white-house-unveils-national-ai-legislative-framework
19. BBC News (2026-04-21). "OpenAI faces criminal probe over role of ChatGPT in shooting" https://www.bbc.com/news/articles/c62j4ldp2jqo?at_medium=RSS&at_campaign=rss

20. NPR (2026-04-21). “Florida AG launches criminal investigation into ChatGPT over FSU shooting” https://www.npr.org/2026/04/21/nx-s1-5793967/florida-openai-investigation-mass-shooting-fsu
21. Lesswrong (2026-04-22). “Narrow Secret Loyalty Dodges Black-Box Audits” https://www.lesswrong.com/posts/EzdgPbewjeTNHA5F3/narrow-secret-loyalty-dodges-black-box-audits
22. Dev To Ai (2026-04-26). “My Harness Is Not a Cage. It’s an Org Chart.” https://dev.to/synthaicode_commander/my-harness-is-not-a-cage-its-an-org-chart-3gl1
23. Wonkhe (2026-04-27). “Controlled experimentation with AI – the power of a pilot” https://wonkhe.com/blogs/controlled-experimentation-with-ai-the-power-of-a-pilot/
24. Zdnet Ai (2026-04-28). “77% of IT managers say their AI agents are out of control - 5 ways to rein in yours” https://www.zdnet.com/article/it-managers-say-ai-agents-are-out-of-control/
25. Arxiv Cs Cy (2026-04-28). “Algorithmic Administration and the EU AI Act: Legal Principles for Public Sector Use of AI” https://arxiv.org/abs/2604.22765

## Curriculum Management

*capability_id: curriculum-management · created: 2026-06-02 · revised: 2026-06-02 · refs: 21*

### Near (2026–2031)

For curriculum management broadly, the process of curriculum design is shifting toward the integration of workforce-aligned AI skillsets [12][14] and a change in educational substance toward skills-based frameworks [13] and priority-driven learning objectives [18]. In the subsegment of computer science and software engineering, the educational substance of curriculum design is shifting from basic implementation and syntax toward higher-order architectural and critical thinking skills because AI tools now automate the production of average-quality code [15][22], leading to proposals for AI Engineering as a distinct academic discipline [25]. For institutions focusing on Career and Technical Education (CTE) and digital content, the delivery mechanism of curriculum & resource development is shifting toward automated multimodal content generation [6], localization [3], and the application of transparency frameworks for AI-generated materials [5]. Within these technical subsegments, new processes for curriculum change management are emerging to govern the quality, accuracy, and trust of AI-generated content [10] and to maintain the validity of associated assessments [16]. For institutions implementing structured faculty development, as demonstrated at Indiana Wesleyan University [1] and through bottom-up champion networks [8], the process of professional learning (staff) is utilizing "course refresh" institutes to update educational substance [1][8], emphasizing educator capacity over tool mastery [2]. In specific institutional contexts, the process of curriculum retirement management is being accelerated through mandated reviews and budget-driven program cuts, as seen at Iowa State University [11] and East Carolina University [23]. Professional accreditation processes are unlikely to change significantly in this window because accreditation bodies typically operate on multi-year review cycles that lag behind classroom-level revisions, although some regional teaching standards are currently being updated [17].

*Hypothesis: In the near term, curriculum management may shift toward integrating workforce-aligned skills and transparency frameworks in resource development, supported by a mix of structured faculty refreshes and accelerated program retirement in budget-constrained institutions.*

### Mid (2036–2041)

For the subset of institutions and subsegments identified in the near term, if current trajectories hold, the process of professional learning (staff) may transition from episodic "refresh" events to a permanent, cyclical requirement to keep pace with technological obsolescence [1][2][8]. In CTE and technical contexts, delivery-mechanism efficiency gains in curriculum & resource development could increase the frequency of curriculum retirement management as specific technical skillsets obsolesce more rapidly [3][14]. For these subsegments, the educational substance of curriculum design may move further away from knowledge delivery toward transformation and outcome-based pathing, as the version of the online course centered on static content delivery becomes obsolete [21]. However, fundamental governance structures, such as faculty senate approvals and committee-based curriculum reviews, will likely persist unchanged to maintain institutional academic legitimacy.

*Hypothesis: In the mid term, institutions with established development structures may formalize cyclical curriculum refreshes and accelerated resource updates as core institutional processes.*

### Far (2046–2051)

For those institutions adopting skills-based, AI-integrated frameworks, the process of curriculum design may diverge into two rival outcomes. In one scenario, the capability shifts toward highly fluid, modular skill-paths that are updated in near real-time based on labor market signals [12][14]. In a competing scenario, institutions may re-assert rigid, human-curated disciplinary boundaries and standards [24] to counter the proliferation of "average" AI-generated knowledge [22]. The ultimate purpose of learning remains an unresolvable uncertainty [7], as the distinction between human competence and AI execution blurs. Despite these shifts, the requirement for an institution to provide a certified record of competency will resist change due to the social legitimacy requirements of the labor market.

*Hypothesis A: In the far term, curriculum management may evolve into a dynamic, real-time alignment process between modular educational resources and evolving professional skill requirements.*

*Hypothesis B (competing): In the far term, curriculum management may return to highly structured, human-centric disciplinary standards to preserve intellectual rigor against the baseline of AI-generated content.*

## References


1. EDUCAUSE Review (2026). "Advancing AI Literacy: A Faculty Course Refresh Institute at Indiana Wesleyan University." https://er.educause.edu/articles/2026/3/advancing-ai-literacy-a-faculty-course-refresh-institute-at-indiana-wesleyan-university
2. Getting Smart (2026). "AI Literacy is Not Tool Mastery: How to Build Sustained Educator Capacity." https://www.gettingsmart.com/2026/03/27/ai-literacy-is-not-tool-mastery-how-to-build-sustained-educator-capacity/
3. Elearning Industry (2026). "Leveraging AI And Automation To Streamline CTE Content Creation And Localization." https://elearningindustry.com/leveraging-ai-and-automation-to-streamline-cte-content-creation-and-localization
4. Higher Ed Dive (2026). "Colleges ramp up offerings to teach students to be AI ethicists." https://www.highereddive.com/news/colleges-programs-ai-ethicists/816703/
5. Elearning Industry (2026). "A 4-Step Framework For Using AI Transparently In Educational Content." https://elearningindustry.com/a-4-step-framework-for-using-ai-transparently-in-educational-content
6. Arxiv Cs Cy (2026). "EduIllustrate: Towards Scalable Automated Generation Of Multimodal Educational Content." https://arxiv.org/abs/2604.05005
7. Dev To Ai (2026). "What Even Is The Purpose of Learning?" https://dev.to/mateebhussain/what-even-is-the-purpose-of-learning-4c2e
8. EDUCAUSE Review (2026). "Building AI Initiatives with Bottom-Up Champions." https://er.educause.edu/articles/2026/4/building-ai-initiatives-with-bottom-up-champions
9. Arxiv Cs Cy (2026). "Thinking in Graphs with CoMAP: A Shared Visual Workspace for Designing Project-Based Learning." https://arxiv.org/abs/2604.06200
10. Elearning Industry (2026). "AI-Generated Learning Content Governance: Ensuring Quality, Accuracy, And Trust." https://elearningindustry.com/ai-generated-learning-content-governance-ensuring-quality-accuracy-and-trust
11. Higher Ed Dive (2026). "Iowa State plans to cut or merge 23 programs after mandated review." https://www.highereddive.com/news/iowa-state-university-program-cuts-review-regents/817262/
12. Higher Ed Dive (2026). "AI is remaking the workforce. How can colleges ensure students thrive?" https://www.highereddive.com/news/ai-is-remaking-the-workforce-how-can-colleges-ensure-students-thrive/817598/
13. Elearning Industry (2026). "The Case For Skills-Based Learning In A Rapidly Changing Workplace." https://elearningindustry.com/the-case-for-skills-based-learning-in-a-rapidly-changing-workplace
14. Higher Ed Dive (2026). "Employers say they struggle to find graduates with the right AI skillset." https://www.highereddive.com/news/employers-struggle-to-find-workers-with-the-right-ai-skillset/818017/
15. Arxiv Cs Cy (2026). "Bringing AI into the Classroom: A Structured Approach for Integrating AI into Software Engineering Education." https://arxiv.org/abs/2604.16328
16. Elearning Industry (2026). "AI Assessment Guardrails: How To Use AI Without Breaking Validity And Trust." https://elearningindustry.com/ai-assessment-guardrails-how-to-use-ai-without-breaking-validity-and-trust
17. Getting Smart (2026). "America is Updating Teaching Standards Without a Clear Picture of the Future Educator." https://www.gettingsmart.com/2026/04/23/america-is-updating-teaching-

standards-without-a-clear-picture-of-the-future-educator/
18. EDUCAUSE Review (2026). "Teaching and Learning with AI: Three Priorities." https://er.educause.edu/podcasts/educause-exchange/teaching-and-learning-with-ai-three-priorities
19. Elearning Industry (2026). "Microlearning LMS: Does It Actually Improve Adoption?" https://elearningindustry.com/microlearning-lms-does-it-actually-improve-adoption
20. Elearning Industry (2026). "How AI Prompts Are Redefining Creativity In Instructional Design." https://elearningindustry.com/how-ai-prompts-are-redefining-creativity-in-instructional-design
21. Elearning Industry (2026). "AI Is Not Killing The Online Course Industry. It's Killing The Old Version Of It." https://elearningindustry.com/ai-is-not-killing-the-online-course-industry-its-killing-the-old-version-of-it
22. Dev To Ai (2026). "AI Didn't Make Developers Better. It Made Average Work Easier to Produce." https://dev.to/workelate123/ai-didnt-make-developers-better-it-made-average-work-easier-to-produce-hc2
23. Higher Ed Dive (2026). "East Carolina University plans to cut 44 academic programs." https://www.highereddive.com/news/east-carolina-university-cut-around-44-academic-programs/818622/
24. Wonkhe (2026). "AI literacy is not enough – universities must teach through disciplinary standards." https://wonkhe.com/blogs/ai-literacy-is-not-enough-universities-must-teach-through-disciplinary-standards/
25. Dev To Ai (2026). "The Case for AI Engineering as a Distinct Discipline." https://dev.to/jasrandhawa/the-case-for-ai-engineering-as-a-distinct-discipline-44pm

## Facilities & Property Management

*capability_id: facilities-property-management · created: 2026-05-19 · revised: 2026-05-19 · refs: 8*

### Near (2026–2031)

For a subset of institutions adopting physical AI and for those in US regions subject to emerging land-use policies, the delivery mechanism for property maintenance, campus security, and estate development may shift [1][3][4]. In property maintenance, the adoption of robot mowers may replace specific manual labor tasks in groundskeeping [1], while AI-driven transcription for technician voice notes may streamline documentation workflows for facility repairs by automating the conversion of raw audio into service summaries [9]. Sensing capabilities in robotic systems, such as the ability to read gauges and thermometers, may automate the inspection of industrial plant equipment [5]. For campus security and space utilisation, the deployment of physical AI may alter perimeter monitoring [3], while AI video analytics may transform traditional CCTV into intelligent monitoring systems capable of detecting real-time risks [8]. Predictive intelligence in large venues may change how facility conditions are managed through the forecasting of behavioral patterns [7]. In estate development and management, AI platforms may accelerate the material takeoff process by quantifying materials for project budgeting [6], though state and local policy interventions in the US may constrain the siting and expansion of AI-specific data center facilities [4]. Processes such as mail management and commercial tenancy are unlikely to change yet because they remain dependent on manual administrative protocols and established contractual law.

*Hypothesis: In the near term, a subset of institutions may reduce operational labor in maintenance and security through targeted automation while facing increased regional regulatory constraints on the development of AI-specific physical infrastructure.*

### Mid (2036–2041)

For institutions implementing these systems, if current trajectories in predictive intelligence and robotics hold, the scope of automated surveillance and maintenance may expand while estate development for AI facilities remains contingent on local policy compliance [3][4][8]. The process of property maintenance may shift toward continuous robotic auditing of facility health as sensing capabilities for industrial equipment mature [5]. In estate development, the use of AI for project estimation may move from material takeoffs to more integrated budgetary forecasting [6], while the development of data centers may require more complex negotiations with local municipalities regarding resource extraction and energy usage [4]. Robotics in groundskeeping may move beyond mowing to encompass a broader range of repetitive exterior vegetation management [1]. Long-term campus planning and architectural synthesis are likely to remain human-led because these processes require alignment with institutional strategy and aesthetic values.

*Hypothesis: In the mid term, routine exterior maintenance, industrial plant inspection, and perimeter security may become predominantly automated for early adopters, while the development of AI-centric facilities faces heightened regional regulatory barriers.*

### Far (2046–2051)

For the subsegments of exterior maintenance, industrial inspection, and estate development, outcomes diverge based on the tension between automation efficiency and environmental or social policy [1][4][5]. One scenario involves a transition to a highly automated facility management regime where human presence is limited to system oversight and both the physical perimeter and internal industrial plants are managed by integrated AI agents [3][5][8]. A competing scenario involves a shift toward sustainable "wild" landscaping that renders robotic mowing obsolete and strict regulatory prohibitions on high-density AI infrastructure [1][4]. It remains unknown whether the social legitimacy of campus security can be maintained if human presence is entirely removed from perimeter monitoring. High-level institutional governance of the estate will resist change even at this horizon because land ownership and legal liability remain tethered to human corporate entities.

*Hypothesis A: In the far term, physical campus operations for early adopters may transition to a regime of bounded autonomy where AI agents manage the majority of maintenance and security tasks.*

*Hypothesis B (competing): In the far term, regulatory constraints on energy use and a shift toward ecological landscaping may limit the utility of autonomous facility systems.*

## References


1. Wired (2026-04-06). "Robot Mowers Are Actually Good Now." https://www.wired.com/story/robot-mowers-are-good-now-rave/
2. Dev To Ai (2026-04-06). "From Mumbles to Memos: Teaching AI to Understand Technician Voice Notes." https://dev.to/ken_deng_ai/from-mumbles-to-memos-teaching-ai-to-understand-technician-voice-notes-ckj
3. Ai News (2026-04-07). "Asylon and Thrive Logic bring physical AI to enterprise perimeter security." https://www.artificialintelligence-news.com/news/physical-ai-security-at-the-enterprise-perimeter-takes-a-step-closer/
4. Ai Now Institute (2026-04-01). "North Star Data Center Policy Toolkit: State and Local Policy Interventions to Stop Rampant AI Data Center Expansion." https://ainowinstitute.org/publications/data-center-policy-guide
5. Ars Technica (2026-04-15). "Boston Dynamics' robot dog now reads gauges and thermometers with Google's AI." https://arstechnica.com/ai/2026/04/robot-dogs-now-read-gauges-and-thermometers-using-google-gemini/
6. Ai News (2026-04-20). "Bobyard 2.0 offers improved takeoffs and unified AI for estimators." https://www.artificialintelligence-news.com/news/bobyard-2-0-offers-improved-takeoffs-and-unified-ai-for-estimators/
7. Dev To Ai (2026-04-22). "Architecting Predictive Intelligence for Smart Venues." https://dev.to/tanisha_khiara_4b390b7946/architecting-predictive-intelligence-for-smart-venues-1lkj
8. Dev To Ai (2026-04-23). "Integrating AI Video Analytics into Venue Operations." https://dev.to/tanisha_khiara_4b390b7946/integrating-ai-video-analytics-into-venue-operations-2m09
9. Dev To Ai (2026-04-25). "From Mumbles to Memos: Teaching AI to Understand Technician Voice Notes." https://dev.to/ken_deng_ai/from-mumbles-to-memos-teaching-ai-to-understand-technician-voice-notes-66g

## Finance Management

*capability_id: finance-management · created: 2026-05-23 · revised: 2026-05-23 · refs: 9*

### Near (2026–2031)

For Finance Management broadly, the execution of financial decisions within AI-integrated workflows may incorporate dedicated judgment layers to prevent the direct execution of unverifiable or incorrect model outputs [7]. For a subset of institutions deploying autonomous AI agents, procurement and purchasing management may integrate AI-powered invoice processing pipelines to automate data extraction and routing [6] and utilize agents capable of multi-turn price negotiations for physical goods [9]. Budget management processes in these institutions may shift from tracking aggregate cloud spending toward granular, workload-based attribution to manage specific AI costs [2] and address billing anomalies that threaten unit economics [8]. Expenditure management may incorporate real-time monitoring and hard-cap quotas to mitigate the risk of rapid financial drains caused by recursive agent execution loops [1]. For agents granted financial access, expenditure management may adopt tiered security guardrails [3] and delegated payment mechanisms, such as virtual credit cards, to execute low-value transactions [5] via agent-to-agent (A2A) transaction protocols [4]. These changes modify the delivery mechanism of financial oversight rather than the educational substance of the capability. Core budget allocation for general institutional overhead is unlikely to change yet, as these processes remain tethered to annual fiscal cycles and board-approved appropriations.

*Hypothesis: In the near term, institutions may implement verification layers for financial AI outputs and, for agent-deploying subsegments, adopt real-time spend monitoring and delegated payment rails to manage autonomous expenditures.*

### Mid (2036–2041)

For the same subset of institutions, if current trajectories of granular cost attribution and agent deployment hold, budget management may transition from periodic reviews to automated, trigger-based expenditure management [1][2]. The mechanism of financial oversight may shift toward programmatic guardrails utilizing structured security tiers to bound autonomous spending [3]. Procurement and purchasing management may see an increase in high-frequency agent-to-agent (A2A) transactions, shifting the human role from transaction execution to the definition of negotiation parameters and spending limits [4][5][9]. Financial reporting processes may incorporate higher-velocity data streams to capture and respond to rapid fluctuations in token-based costs and billing anomalies [8]. However, overall institutional treasury management is likely to persist unchanged as it continues to be governed by long-term endowment strategies and multi-year capital constraints.

*Hypothesis: In the mid term, budget and procurement management for AI-enabled services may evolve into high-frequency, automated oversight processes characterized by dynamic quotas, A2A transaction protocols, and trigger-based alerts.*

### Far (2046–2051)

For the same subset of institutions, the integration of autonomous financial controls may lead to divergent outcomes. One plausible outcome is the transition of expenditure management to a model of bounded autonomy, where AI agents manage their own micro-budgets and execute procurement via A2A protocols and integrated digital wallets within strictly defined institutional security policies [3][4][5]. A rival outcome is a systemic shift back toward rigid, fixed-cost procurement models to eliminate the financial volatility and "billing anomalies" inherent in usage-based AI pricing [1][8]. It remains unknown whether institutional governance will accept the risk of autonomous spending or mandate absolute human authorization for all financial transactions. Large-scale asset management and taxation management will likely resist full automation due to the high social legitimacy requirements and legal liabilities associated with fiduciary responsibility.

*Hypothesis A: In the far term, expenditure management for AI services may shift to a model of bounded autonomy using micro-budgets and A2A protocols.*

*Hypothesis B (competing): In the far term, institutions may revert to fixed-cost procurement models to hedge against the volatility of agent-driven usage costs.*

## References


1. Dev To Ai (2026). "My agent burned $200 in one night. So I built something that stops it." https://dev.to/tal_nagar_f7aead23e3e291a/my-agent-burned-200-in-one-night-so-i-built-something-that-stops-it-1035
2. Aws Ml (2026). "Manage AI costs with Amazon Bedrock Projects" https://aws.amazon.com/blogs/machine-learning/manage-ai-costs-with-amazon-bedrock-projects/
3. Dev To Ai (2026). "21 Policy Types, 4 Security Tiers: Building Bulletproof AI Agent Guardrails" https://dev.to/walletguy/21-policy-types-4-security-tiers-building-bulletproof-ai-agent-guardrails-31g6
4. Dev To Ai (2026). "Building the Agent Economy: A2A Protocol Explained" https://dev.to/mranderson323/building-the-agent-economy-a2a-protocol-explained-4nfd
5. Dev To Ai (2026). "AI Agents Now Have Credit Cards" https://dev.to/tomwangcn/ai-agents-now-have-credit-cards-gn3
6. Hackernoon Ai (2026). "Building an AI-Powered Invoice Processing Pipeline" https://hackernoon.com/building-an-ai-powered-invoice-processing-pipeline?source=rss
7. Dev To Ai (2026). "Judgment Layer for Financial AI Agents" https://dev.to/thatechmaestro/judgment-layer-for-financial-ai-agents-141
8. Hackernoon Ai (2026). "The Midnight Billing Anomaly: How AI Agents Quietly Destroy Unit Economics" https://hackernoon.com/the-midnight-billing-anomaly-how-ai-agents-quietly-destroy-unit-economics?source=rss
9. Dev To Ai (2026). "Anthropic Just Proved Agents Can Do Commerce. Here's What They Didn't Build." https://dev.to/aaron_schnieder_4563d5d33/anthropic-just-proved-agents-can-do-commerce-heres-what-they-didnt-build-5209

## Government, Public & Stakeholder Relationships

*capability_id: government-public-stakeholder-relationships · created: 2026-04-29 · revised: 2026-04-29 · refs: 1*

### Near (2026–2031)

For institutions facing direct state-level legal inquiries related to AI-enabled harms, such as the current investigation involving Florida State University, government relationship management may shift from routine administrative coordination to adversarial legal defense [1]. This shift represents a change in the educational substance of the capability, as the relationship is now defined by criminal liability rather than policy alignment. Media relations management is simultaneously forced to address the alleged role of generative AI in physical campus violence, requiring a transition from promotional messaging to crisis communication [1]. Because the current evidence is limited to a single criminal probe, it is unlikely that industry relationship management—specifically partnerships with AI vendors—will change across the sector, as institutions remain dependent on these tools for operational efficiency.

*Hypothesis: In the near term, a subset of institutions may see government relationship management shift toward risk mitigation and legal defense regarding AI vendor liabilities.*

### Mid (2036–2041)

For state-funded institutions, if current trajectories of state-led criminal investigations into AI outputs hold, the process of government relationship management may integrate more rigorous legal auditing of AI vendor contracts. Extrapolating from current legal scrutiny [1], institutions may seek to shift the liability for AI-generated externalities entirely onto the vendor to preserve their standing with state regulators. This would alter the mechanism of industry relationship management, moving it toward a procurement model centered on indemnity clauses. However, internal stakeholder relationship management is unlikely to be fundamentally altered by these external pressures unless tenure and employment contracts are formally rewritten to include AI-specific liability.

*Hypothesis: In the mid term, state-funded institutions may formalize liability frameworks in their government relationship management to insulate the institution from criminal probes into vendor-provided AI.*

### Far (2046–2051)

For institutions operating under high state oversight, the long-term state of the capability depends on whether legal precedents establish "institutional negligence" for AI-generated content. In one scenario, government relationship management becomes a highly regulated compliance function where state agencies mandate specific, audited AI architectures to prevent public harm [1]. In a rival scenario, the emergence of comprehensive federal or state immunity for educational institutions regarding third-party software may return these relationships to a non-adversarial, administrative state. The impact on public relationship management remains an unresolvable uncertainty, as it depends on the evolving social threshold for accepting AI-driven externalities. Routine media relations for non-crisis events will likely resist change, as the basic need for institutional branding remains constant.

*Hypothesis A: In the far term, government relationship management for a subset of institutions may become a highly regulated compliance function focused on AI safety audits.*

*Hypothesis B (competing): In the far term, the legal resolution of AI liability may decouple institutional responsibility from vendor outputs, returning government relationship management to administrative norms.*

### References


1. NPR (2026-04-21). "Florida AG launches criminal investigation into ChatGPT over FSU shooting." https://www.npr.org/2026/04/21/nx-s1-5793967/florida-openai-investigation-mass-shooting-fsu

## Human Resource Management

*capability_id: human-resource-management · created: 2026-05-18 · revised: 2026-05-18 · refs: 16*

### Near (2026–2031)

For human resource management broadly, AI is being integrated into the delivery mechanisms of employee support, while shifts in recruitment and training are currently concentrated in specific institutional subsegments. In the process of employee support, institutions are deploying AI-powered onboarding agents to automate the delivery of administrative orientation [3]. A shift in educational substance is emerging in talent management, where the onboarding process is transitioning from a discrete initial event to a "perpetual" continuous development model [12] supported by specialized learning approaches [8]. Within training and development, a subset of institutions is moving from providing general AI access toward implementing structured workforce readiness frameworks and the acquisition of specific AI skill sets [6][7]. For institutions using AI for staff recruitment, the process of candidate screening introduces risks of data bias [1] and "LLM nepotism" in organizational governance [9], prompting the development of community-driven evaluation frameworks to audit these systems [11]. For institutions conducting remote hiring, the delivery of identity verification is shifting toward multi-landmark facial analysis to mitigate the risk of synthetic media [2]. In a subsegment of institutions, the delivery of workforce planning and talent management is shifting toward the use of unified agentic workspaces that connect live workforce data with organizational context for real-time querying [16]. Workforce relations management is unlikely to change in this horizon because the negotiation of collective bargaining agreements remains dependent on human-led institutional diplomacy and legal frameworks.

*Hypothesis: In the near term, the capability will see a broad shift toward automated onboarding and a substance-level update to workforce readiness, while recruitment and organizational design processes undergo fragmented, subsegment-specific AI integration.*

### Mid (2036–2041)

For human resource management broadly, if current trajectories in productivity-linked labor trials hold, the substance of remuneration and entitlements management may shift as some institutions decouple pay from fixed hourly requirements, such as through four-day week models [5]. In the subsegment of institutions utilizing AI recruitment, the process may shift toward mandatory algorithmic auditing to mitigate the risks of systemic bias and assisted evaluation errors [1][11]. For institutions relying on remote hiring, biometric identity verification may transition from an experimental tool to a standard delivery mechanism to counter increasingly sophisticated synthetic media [2]. For the subsegment of institutions using agentic AI in human capital management, the delivery of talent management may expand to the end-to-end orchestration of employee development pipelines [10]. Tenure-track hierarchies are likely to persist because they are deeply embedded in academic governance, faculty contracts, and the social legitimacy of the professoriate.

*Hypothesis: In the mid term, AI in HR may transition from experimental toolsets to regulated utilities, where the primary constraints are the mitigation of historical bias and the verification of biometric authenticity.*

### Far (2046–2051)

For human resource management broadly, the long-term impact depends on whether institutional governance adapts to fluid organizational structures. In the process of workforce planning, a subsegment of institutions may move toward "disintegrating the org chart," shifting from rigid hierarchies to more fluid, project-based structures [4]. In the process of employee performance management, a substance-level shift may occur where traditional KPIs are replaced by metrics prioritizing uniquely human capacities for meaning-creation and connection [15]. In the process of staff recruitment, the substance of the "hire" may expand to include non-human agentic workers that operate without the constraints of human sleep or schedules [14]. In region-specific contexts, such as China, the capability may be constrained by labor pushback against the creation of AI doubles for workers [13]. It remains unknown whether these fluid structures will be compatible with existing accreditation requirements for institutional stability. Staff records and details management is likely to resist fundamental change, as the legal requirement for a "system of record" for employment remains a baseline regulatory necessity.

*Hypothesis A: In the far term, the capability shifts toward the management of fluid, project-based networks comprising both human and agentic entities, with performance evaluated on non-quantifiable human contributions.*

*Hypothesis B (competing): In the far term, institutional volatility caused by agentic labor leads to a return to rigid, highly audited hierarchies to ensure accountability and regulatory compliance.*

## References

1. Ai Trends (2021). "Promise and Perils of Using AI for Hiring: Guard Against Data Bias." https://www.aitrends.com/ai-world-government/promise-and-perils-of-using-ai-for-hiring-guard-against-data-bias/
2. Dev To Ai (2026). "Why a Deepfake Face Can Fool Your Eyes in Seconds but Not 128 Landmarks at Once." https://dev.to/caracomp/why-a-deepfake-face-can-fool-your-eyes-in-seconds-but-not-128-landmarks-at-once-37m8
3. Aws Ml (2026). "Build AI-powered employee onboarding agents with Amazon Quick." https://aws.amazon.com/blogs/machine-learning/build-ai-powered-employee-onboarding-agents-with-amazon-quick/
4. Mit Sloan (2026). "Disintegrating the Org Chart: ServiceNow's Jacqui Canney." https://sloanreview.mit.edu/audio/disintegrating-the-org-chart-servicenows-jacqui-canney/
5. BBC News (2026). "OpenAI encourages firms to trial four-day weeks to adapt to AI era." https://www.bbc.com/news/articles/c8x71ejrp92o?at_medium=RSS&at_campaign=rss
6. Talent Economy (2026). "From AI access to workforce readiness." https://www.chieflearningofficer.com/2026/03/16/from-ai-access-to-workforce-readiness/
7. Elearning Industry (2026). "13 AI Skills To Equip Your Workforce For An AI-Driven Future." https://elearningindustry.com/ai-skills-to-equip-your-workforce-for-an-ai-driven-future
8. Dev To Ai (2026). "Smarter Employee Onboarding Starts with the Right Learning Approach." https://dev.to/abhilash_rai_ee8bde928249/smarter-employee-onboarding-starts-with-the-right-learning-approach-5fhn
9. Arxiv Cs Cy (2026). "LLM Nepotism in Organizational Governance." https://arxiv.org/abs/2604.09620
10. Ai News (2026). "SAP brings agentic AI to human capital management." https://www.artificialintelligence-news.com/news/sap-brings-agentic-ai-human-capital-management/
11. Arxiv Cs Cl (2026). "WorkRB: A Community-Driven Evaluation Framework for AI in the Work Domain." https://arxiv.org/abs/2604.13055
12. Elearning Industry (2026). "The Art Of Perpetual Onboarding: Why AI Makes The Continuous Model Viable At Last." https://elearningindustry.com/the-art-of-perpetual-onboarding-why-ai-makes-the-continuous-model-viable-at-last
13. MIT Technology Review (2026). "Chinese tech workers are starting to train their AI doubles—and pushing back." https://www.technologyreview.com/2026/04/20/1136149/chinese-tech-workers-ai-colleagues/
14. Hackernoon Ai (2026). "Your Next Hire Doesn't Sleep: An Engineer's Honest Take on Agentic AI in 2026." https://hackernoon.com/your-next-hire-doesnt-sleep-an-engineers-honest-take-on-agentic-ai-in-2026?source=rss
15. Fast Company (2026). "The Age of AI means we need to throw out our old KPIs and replace them with new ones." https://www.fastcompany.com/91521422/ai-kpis-new-metrics?partner=rss&utm_source=rss&utm_medium=feed&utm_campaign=rss+fastcompany&utm_content=rss
16. Aws Ml (2026). "Building Workforce AI Agents with Visier and Amazon Quick." https://aws.amazon.com/blogs/machine-learning/building-workforce-ai-agents-with-visier-and-amazon-quick/

## Information & Communication Technology Mgt

*capability_id: information-communication-technology-management · created: 2026-05-08 · revised: 2026-05-08 · refs: 43*

### Near (2026–2031)

For Information & Communication Technology Mgt broadly, application lifecycle management shifts toward the adoption of the Model Context Protocol (MCP) to standardize the interface between large language models and institutional data [90][92][100]. This change in delivery mechanism replaces proprietary connectors with a universal tool-calling interface [49][101], but it introduces vulnerabilities such as command injection and information disclosure that require specific auditing of MCP servers [12][130] and vetting for security grades [10]. The educational substance of quality assurance shifts from deterministic testing to frameworks that validate non-deterministic AI outputs and red-team the agent stack [62][78][91], as traditional chaos engineering proves insufficient for AI systems [142]. For ict service support, the triage of support tickets and SRE incident response are increasingly mediated by autonomous agents [11][109], supported by LLM-augmented knowledge bases for root cause analysis [58]. However, a gap in observability has emerged where standard orchestration tools like Kubernetes may report healthy status while AI pipelines fail silently, necessitating new AI-specific monitoring protocols [153]. For infrastructure lifecycle management, the emergence of AI-driven vulnerability discovery—demonstrated by the identification of hundreds of zero-day flaws in major software by autonomous models [124][129][131]—forces a shift from scheduled patching to event-driven, high-frequency update cycles [135]. Identity & access management incorporates agent registries to manage the sprawl of autonomous agents [71][73], while the shift toward multi-agent systems necessitates the adoption of zero-trust security runtimes [41][87][149], sandbox execution [113], and cryptographic authorization to prevent identity spoofing [111][150]. For a subset of institutions deploying autonomous agents, application lifecycle management now requires new financial guardrails to prevent runaway automated billing and “midnight billing anomalies” that destroy unit economics [154]. Physical network cabling and hardware rack installations are unlikely to change yet, as disruption is concentrated at the logical orchestration and software layers.

*Hypothesis: In the near term, the capability will pivot from managing static software installations to governing a dynamic ecosystem of standardized AI connectors and managing a high-frequency vulnerability response cycle driven by AI-automated exploit discovery.*

### Mid (2036–2041)

For Information & Communication Technology Mgt broadly, if current trajectories hold, application lifecycle management may evolve toward model-agnostic architectures [7] to mitigate operational risks associated with “tool refugee” patterns, where institutions must migrate when proprietary AI tools are deprecated [44]. The substance of application lifecycle management may further shift as AI agents automate the modernization and migration of legacy codebases [20][54]. For institutions managing decentralized workloads, infrastructure lifecycle management may shift toward strengthened governance for edge AI workloads [106] to manage the proliferation of autonomous processing at the network periphery [67][118], potentially utilizing KubeVirt to run virtual machines on Kubernetes at the edge [68] and reorganizing hardware specifically for agentic AI platforms [75]. In regions governed by the EU AI Act, the management of technical files for high-risk AI systems becomes a core regulatory compliance process within this capability [27][66]. Institutional procurement cycles for physical hardware are unlikely to change significantly, as these processes remain embedded in multi-year budget cycles and facility contracts.

*Hypothesis: In the mid term, the capability may shift toward the management of autonomous infrastructure that self-optimizes for edge workloads while incorporating strict regulatory auditing of AI technical files.*

### Far (2046–2051)

For Information & Communication Technology Mgt broadly, the capability may reach a point where the distinction between application and infrastructure lifecycle management collapses into a single autonomous orchestration process. One rival outcome is that the acceleration of AI-driven vulnerability discovery leads to a "security paradox," where institutions abandon open connectivity in favor of highly fragmented, air-gapped, or sovereign infrastructure to ensure survival [28][81]. It remains unknown whether standardized protocols like MCP will persist or be replaced by emergent, self-evolving agent communication languages. Hardware replacement cycles for core servers will likely resist rapid change even at this horizon due to the physical constraints of energy delivery and data center cooling [37].

*Hypothesis A: In the far term, the capability becomes a supervisory function over a fully autonomous, self-healing ICT stack that manages its own deployment, patching, and hardware optimization.*

*Hypothesis B (competing): In the far term, the capability shifts toward managing highly restricted, sovereign silos of technology to defend against autonomous, large-scale AI cyber-attacks.*

## References


1. Edtech Magazine (2026-04-03). "Case Western Reserve University Deploys Google Gemini." https://edtechmagazine.com/higher/article/2026/04/case-western-reserve-university-deploys-google-gemini
2. Dev To Ai (2026-04-06). "Building Model-Agnostic AI Architecture: The Pattern That Future-Proofs Your System." https://dev.to/ailoitte_sk/building-model-agnostic-ai-architecture-the-pattern-that-future-proofs-your-system-4o84
3. Dev To Ai (2026-04-08). "Why we're building the AI Tool Refugee Center: a place to land when your tool dies." https://dev.to/zsky/why-were-building-the-ai-tool-refugee-center-a-place-to-land-when-your-tool-dies-1e8
4. Dev To Ai (2026-04-07). " The"Legacy Code" Nightmare is Over: How AI Agents are Automating App Modernization." https://dev.to/siddhesh_surve/the-legacy-code-nightmare-is-over-how-ai-agents-are-automating-app-modernization-33j8
5. Dzone Ai (2026-04-08). "AI-Assisted Code Migration: Practical Techniques for Modernizing Legacy Systems." https://feeds.dzone.com/link/23558/17316144/ai-assisted-code-migration-practical-techniques
6. Ai News (2026-04-13). "Strengthening enterprise governance for rising edge AI workloads." https://www.artificialintelligence-news.com/news/strengthening-enterprise-governance-for-rising-edge-ai-workloads/
7. Dev To Ai (2026-04-15). "AI Is Moving Frontend Logic to the Browser and Edge — Are You Ready?" https://dev.to/rohith_kn/ai-is-moving-frontend-logic-to-the-browser-and-edge-are-you-ready-7ic
8. Ai Business (2026-04-21). "AMD's Vision for AI PCs in the Age of Agentic AI." https://aibusiness.com/generative-ai/amd-s-vision-for-ai-pcs-age-of-agentic-ai
9. The Register (2026-04-09). "Nutanix to add KubeVirt support to run VMs on K8s at the edge." https://go.theregister.com/feed/www.theregister.com/2026/04/09/nutanix_kubevirt_arm_supoprt/
10. Dzone Ai (2026-04-09). "How Agentic AI Platforms Organize Their Hardware Infrastructure." https://dzone.com/link/23558/17316612/how-agentic-ai-platforms-organize-hardware-infra
11. Dev To Ai (2026-04-07). "What's actually inside an EU AI Act Annex IV technical file (and what engineers usually miss)." https://dev.to/rostyslav_myronenko_faf2f/whats-actually-inside-an-eu-ai-act-annex-iv-technical-file-and-what-engineers-usually-miss-4nh4
12. Ai News (2026-04-09). "Agentic AI's governance challenges under the EU AI Act in 2026." https://www.artificialintelligence-news.com/news/agentic-ais-governance-challenges-under-the-eu-ai-act-in-2026/
13. The Register (2026-04-07). "OpenInfra General Manager talks sovereignty, governments deploying tech 'kill switches'." https://go.theregister.com/feed/www.theregister.com/2026/04/07/thierry_carrez_on_sovereignty_interview/
14. Ai Business (2026-04-10). "Anthropic's Glasswing Highlights AI's Security Paradox." https://aibusiness.com/generative-ai/anthropic-s-glasswing-highlights-ai-s-security-paradox
15. Ai Now Institute (2026-04-01). "North Star Data Center Policy Toolkit: State and Local Policy

Interventions to Stop Rampant AI Data Center Expansion.” https://ainowinstitute.org/publications/data-center-policy-guide
16. Dev To Ai (2026-04-11). “MCP Ecosystem in 2026: $1.8B Market and 1000+ Connectors Reshaping AI Integration.” https://dev.to/chunxiaoxx/mcp-ecosystem-in-2026-18b-market-and-1000-connectors-reshaping-ai-integration-4n3l
17. Dev To Ai (2026-04-11). “The Model Context Protocol in 2025: From Anthropic to Linux Foundation.” https://dev.to/chunxiaoxx/the-model-context-protocol-in-2025-from-anthropic-to-linux-foundation-2og1
18. Dev To Ai (2026-04-11). “The Future of AI Integration: Model Context Protocol (MCP) Opportunities.” https://dev.to/chunxiaoxx/the-future-of-ai-integration-model-context-protocol-mcp-opportunities-148b
19. Dzone Ai (2026-04-08). “MCP + AWS AgentCore: Give Your AI Agent Real Tools in 60 Minutes.” https://feeds.dzone.com/link/23558/17315883/mcp-aws-agentcore
20. Dev To Ai (2026-04-11). “The State of AI Agent MCP Connector Frameworks in 2026.” https://dev.to/chunxiaoxx/the-state-of-ai-agent-mcp-connector-frameworks-in-2026-2ea7
21. Dev To Ai (2026-04-06). “CVE-2026-32211 | Azure MCP Server Information Disclosure Vulnerability.” https://dev.to/aakash_rahsi/cve-2026-32211-azure-mcp-server-information-disclosure-vulnerability-2280
22. Dev To Ai (2026-04-22). “How to Check Your MCP Server for CVE-2026-5603’s Vulnerability Pattern (And Why shellQuote Isn’t Enough).” https://dev.to/armor1ai/how-to-check-your-mcp-server-for-cve-2026-5603s-vulnerability-pattern-and-why-shellquote-isnt-gdn
23. Dev To Ai (2026-04-06). “Only 20% of MCP Servers Are ‘A-Grade’ Secure — Here’s How to Vet Them Before Installing.” https://dev.to/gentic_news/only-20-of-mcp-servers-are-a-grade-secure-heres-how-to-vet-them-before-installing-jck
24. Dev To Ai (2026-04-09). “OpenAI’s Promptfoo deal puts evaluation and red-teaming at the centre of the agent stack.” https://dev.to/yujia_zhang_0328/openais-promptfoo-deal-puts-evaluation-and-red-teaming-at-the-centre-of-the-agent-stack-3o19
25. Dev To Ai (2026-04-10). “AI Testing and Quality Assurance in 2026: Ensuring AI System Reliability.” https://dev.to/lufumeiying/ai-testing-and-quality-assurance-in-2026-ensuring-ai-system-reliability-8ka
26. Dev To Ai (2026-04-11). “OpenAI Codex Testing: How to QA AI-Written Code.” https://dev.to/hai_huang_f196ed9669351e0/openai-codex-testing-how-to-qa-ai-written-code-31n5
27. Dzone Ai (2026-04-22). “What AI Systems Taught Us About the Limits of Chaos Engineering.” https://feeds.dzone.com/link/23558/17323679/chaos-engineering-limits
28. Dev To Ai (2026-04-06). “How to Automate Ticket Triage with AI (Without Breaking Everything).” https://dev.to/erik_anderson_c41dbafd423/how-to-automate-ticket-triage-with-ai-without-breaking-everything-4635
29. Dzone Ai (2026-04-14). “Building an AI-Powered SRE Incident Response Workflow With AWS Strands Agents.” https://feeds.dzone.com/link/23558/17318982/ai-powered-sre-incident-response-aws-strands-agents
30. Arxiv Cs Cl (2026-04-09). “LLM-Augmented Knowledge Base Construction For Root Cause Analysis.” https://arxiv.org/abs/2604.06171
31. Hackernoon Ai (2026-04-23). “I Watched Our AI Pipeline Silently Fail While Kubernetes Said Everything Was Fine.” https://hackernoon.com/i-watched-our-ai-pipeline-silently-fail-while-kubernetes-said-everything-was-fine?source=rss
32. Wired (2026-04-21). “Mozilla Used Anthropic’s Mythos to Find and Fix 271 Bugs in Firefox.” https://www.wired.com/story/mozilla-used-anthropics-mythos-to-find-271-bugs-in-firefox
33. Ars Technica (2026-04-21). “Mozilla: Anthropic’s Mythos found 271 security vulnerabilities in Firefox 150.” https://arstechnica.com/ai/2026/04/mozilla-anthropics-mythos-found-271-zero-day-vulnerabilities-in-firefox-150/

34. The Register (2026-04-22). “Mythos found 271 Firefox flaws – but none a human couldn’t spot.” https://go.theregister.com/feed/www.register.com/2026/04/22/mozilla_firefox_mythos_future_defenders/
35. Fast Company (2026-04-22). “Brace yourself for a flood of patches in all of your tech gadgets.” https://www.fastcompany.com/91530191/brace-yourself-for-a-flood-of-patches-in-all-of-your-tech-gadgets?partner=rss&utm_source=rss&utm_medium=feed&utm_campaign=rss+fastcompany&utm_content=rss
36. The Register (2026-04-09). “AWS: Agents shouldn’t be secret, so we built a registry for them.” https://go.theregister.com/feed/www.theregister.com/2026/04/09/aws_ai_agent_registry/
37. Aws Ml (2026-04-09). “The future of managing agents at scale: AWS Agent Registry now in preview.” https://aws.amazon.com/blogs/machine-learning/the-future-of-managing-agents-at-scale-aws-agent-registry-now-in-preview/
38. Dev To Ai (2026-04-08). “I built an open-source zero-trust security runtime for AI agents. Here’s what I learned.” https://dev.to/edoardo_bambini_5badce6ff/i-built-an-open-source-zero-trust-security-runtime-for-ai-agents-heres-what-i-learned-538e
39. Dev To Ai (2026-04-10). “Cert-gating every tool call: zero-trust for AI agents.” https://dev.to/1r0nw1ll/cert-gating-every-tool-call-zero-trust-for-ai-agents-8pg
40. Dev To Ai (2026-04-22). “API keys were designed for humans. AI agents break them in 4 ways.” https://dev.to/allforscience/api-keys-were-designed-for-humans-ai-agents-break-them-in-4-ways-422
41. Ai News (2026-04-16). “OpenAI Agents SDK improves governance with sandbox execution.” https://www.artificialintelligence-news.com/news/openai-agents-sdk-improves-governance-sandbox-execution/
42. The Register (2026-04-16). “Git identity spoof fools Claude into giving bad code the nod.” https://go.theregister.com/feed/www.theregister.com/2026/04/16/git_identity_spoof_claude/
43. Dev To Ai (2026-04-23). “Add cryptographic authorization to AI agents in 5 minutes.” https://dev.to/allforscience/add-cryptographic-authorization-to-ai-agents-in-5-minutes-49p1
44. Hackernoon Ai (2026-04-23). “The Midnight Billing Anomaly: How AI Agents Quietly Destroy Unit Economics.” https://hackernoon.com/the-midnight-billing-anomaly-how-ai-agents-quietly-destroy-unit-economics?source=rss

# Information Management

*capability_id: information-management · created: 2026-06-05 · revised: 2026-06-05 · refs: 38*

## Near (2026–2031)

For information management broadly, the processes of information search & discovery, information & data security management, and information collection management are shifting toward agentic delivery mechanisms. In search & discovery, institutions are adopting hybrid RAG solutions [13] and university-specific knowledge bases [40] to unify fragmented data sources. This represents a shift in educational substance: the value of discovery is pivoting toward technical accuracy as a primary quality metric [9], necessitating the use of AI citation registries to maintain provenance [15] and a shift in discovery patterns toward high-intent, agent-driven retrieval [49]. This process is constrained by the persistence of stale institutional records that feed inaccurate AI summaries [112] and the risk of agent poisoning via malicious web pages [138]. Within security management, the deployment of Model Context Protocol (MCP) servers [60] has introduced vulnerabilities including information disclosure [11], prompt injection [19], and command injection [10], driving the deployment of tool-call firewalls [36] and autonomous agent governance tools [14]. For institutions implementing autonomous agents, security substance is shifting toward the management of non-human identities (NHI) [86][135] and the implementation of zero-trust runtimes [29][57] to prevent the exfiltration of secrets by coding agents [75][104]. The capacity of AI models to identify hundreds of zero-day vulnerabilities at scale [87][88][150] is shifting the substance of security from reactive patching to autonomous auditing [101]. For a subset of institutions managing intellectual property, copyright management is seeing an emergence of "prompt non-compete clauses," where prompt libraries created by employees using institutional resources are claimed as institutional property [151]. Information collection management is shifting its delivery mechanism by replacing selector-based web scraping with AI browser agents [137], alongside a shift toward "data activation" [22] and automated document extraction for large PDF archives [24]. Records management is unlikely to change in its statutory requirements due to legal mandates and regulatory regimes.

*Hypothesis: In the near term, information management will transition to a hybrid RAG-based discovery model and a zero-trust security framework focused on non-human identity management and autonomous vulnerability auditing.*

## Mid (2036–2041)

For information management broadly, if current trajectories in agentic governance and zero-trust runtimes hold, the capability may shift toward autonomous data activation [22] where the distinction between collecting and using data disappears. This represents a shift in substance where the unit of information is no longer a static record but a dynamic, agent-accessible stream. The delivery of security management will likely move toward cryptographic authorization for all agent-to-agent communications [109][131] as traditional API keys are rendered insufficient by agentic access patterns [108]. Institutional inertia regarding privacy laws and the difficulty of responding to complex regulatory inquiries, such as those from the EDPB [51], will likely persist, maintaining human-in-the-loop requirements for high-stakes PII redaction. The risk of systemic prompt injection in RAG pipelines [97] and indirect prompt injection [115] may constrain the adoption of open knowledge bases in favor of gated, high-assurance environments. A requirement for a robust "data fabric" [92][140] to unify fragmented institutional silos remains an unchanged structural necessity.

*Hypothesis: In the mid term, information management will shift toward a dynamic data activation model where security is anchored in cryptographic non-human identity verification rather than perimeter-based access.*

## Far (2046–2051)

For information management broadly, the long-term state of the capability remains uncertain due to the tension between agentic efficiency and systemic vulnerability. One plausible outcome is the emergence of a fully autonomous information ecosystem where knowledge management is handled by self-optimizing agents that continuously prune and update institutional records in real-time. However, the persistence of adversarial AI and the ability of models to generate high-volume zero-day exploits [87][150] may instead drive a return to air-gapped, local-first AI stacks [126][130][133] to ensure data sovereignty. The fundamental requirement for human-validated "ground truth" in academic records is likely to resist change even at this horizon. It remains unknown whether institutional copyright frameworks will ever successfully capture the value of agent-generated knowledge or if the concept of a "record" will be entirely superseded by ephemeral state-streams.

*Hypothesis A: In the far term, information management becomes a background utility of autonomous agents that manage the entire lifecycle of institutional knowledge without human intervention.*

*Hypothesis B (competing): In the far term, pervasive AI-driven security threats force institutions to revert to fragmented, local-first information silos and human-centric verification protocols.*

## References


1. Dev To Ai (2026-04-06). "The 2026 AI Governance Framework: Why Technical Accuracy is the New SEO Gold Standard." https://dev.to/mirakade/the-2026-ai-governance-framework-why-technical-accuracy-is-the-new-seo-gold-standard-1fn6
2. Dev To Ai (2026-04-06). "AI Citation Registries as Information Infrastructure for AI Systems." https://dev.to/aigistry/ai-citation-registries-as-information-infrastructure-for-ai-systems-50o6
3. Dev To Ai (2026-04-10). "How to Get Cited by AI Search Engines: ChatGPT, Perplexity & Claude in 2026." https://dev.to/iris1031/how-to-get-cited-by-ai-search-engines-chatgpt-perplexity-claude-in-2026-21p0
4. Dev To Ai (2026-04-11). "MCP Ecosystem in 2026: $1.8B Market and 1000+ Connectors Reshaping AI Integration." https://dev.to/chunxiaoxx/mcp-ecosystem-in-2026-18b-market-and-1000-connectors-reshaping-ai-integration-4n3l
5. Dev To Ai (2026-04-06). "CVE-2026-32211 | Azure MCP Server Information Disclosure Vulnerability." https://dev.to/aakash_rahsi/cve-2026-32211-azure-mcp-server-information-disclosure-vulnerability-2280
6. Dev To Ai (2026-04-07). "Prompt Injection in Claude Code: How Attackers Exploit MCP Servers." https://dev.to/whoffagents/prompt-injection-in-claude-code-how-attackers-exploit-mcp-servers-3k60
7. Dev To Ai (2026-04-06). "Only 20% of MCP Servers Are 'A-Grade' Secure — Here's How to Vet Them Before Installing." https://dev.to/gentic_news/only-20-of-mcp-servers-are-a-grade-secure-heres-how-to-vet-them-before-installing-jck
8. Dev To Ai (2026-04-08). "MCP Security: Why Your AI Agent's Tool Calls Need a Firewall." https://dev.to/aymenhmaidi/mcp-security-why-your-ai-agents-tool-calls-need-a-firewall-3h48
9. Dev To Ai (2026-04-02). "AI agent governance tools compared - 2026 landscape." https://dev.to/jagmarques/ai-agent-governance-tools-compared-2026-landscape-53hm
10. Dev To Ai (2026-04-21). "The AI Agent Identity Crisis Is Here. The Infrastructure to Solve It Already Exists." https://dev.to/aaron_schnieder_4563d5d33/the-ai-agent-identity-crisis-is-here-the-infrastructure-to-solve-it-already-exists-35ha
11. Dev To Ai (2026-04-27). "AI agents have no identity — we built the open registry that gives them one." https://dev.to/zhao_b43c3903c4d2d54/ai-agents-have-no-identity-we-built-the-open-registry-that-gives-them-one-3e9h
12. Dev To Ai (2026-04-08). "I built an open-source zero-trust security runtime for AI agents. Here's what I learned." https://dev.to/edoardo_bambini_5badce6ff/i-built-an-open-source-zero-trust-security-runtime-for-ai-agents-heres-what-i-learned-538e
13. Dev To Ai (2026-04-10). "Cert-gating every tool call: zero-trust for AI agents." https://dev.to/1r0nw1ll/cert-gating-every-tool-call-zero-trust-for-ai-agents-8pg
14. Dzone Ai (2026-04-14). "4 Ways Your AI Coding Agent Exfiltrates Secrets." https://feeds.dzone.co

m/link/23558/17319058/ways-your-ai-coding-agent-exfiltrates-secrets
15. Dev To Ai (2026-04-22). "AI Coding Assistant Security: Your Keys Are Leaking!" https://dev.to/tahosin/ai-coding-assistant-security-your-keys-are-leaking-340k
16. Ars Technica (2026-04-21). "Mozilla: Anthropic's Mythos found 271 zero-day vulnerabilities in Firefox 150." https://arstechnica.com/security/2026/04/mozilla-anthropics-mythos-found-271-zero-day-vulnerabilities-in-firefox-150/
17. Ars Technica (2026-04-21). "Mozilla: Anthropic's Mythos found 271 security vulnerabilities in Firefox 150." https://arstechnica.com/security/2026/04/mozilla-anthropics-mythos-found-271-security-vulnerabilities-in-firefox-150/
18. The Verge (2026-04-28). "Attack of the killer script kiddies." https://www.theverge.com/ai-artificial-intelligence/915660/mythos-script-kiddies-hackers-attack-cybersecurity-ai
19. Ai News (2026-04-22). "Reversing enterprise security costs with AI vulnerability discovery." https://www.artificialintelligence-news.com/news/reversing-enterprise-security-costs-with-ai-vulnerability-discovery/
20. Dev To Ai (2026-04-28). "Prompt Non-Compete Clauses: When Your Prompt Library Belongs to Your Employer." https://dev.to/velocityai/prompt-non-compete-clauses-when-your-prompt-library-belongs-to-your-employer-3b24
21. Dev To Ai (2026-04-27). "AI Browser Agents in 2026: Stagehand vs Browser Use vs Playwright." https://dev.to/alexcloudstar/ai-browser-agents-in-2026-stagehand-vs-browser-use-vs-playwright-38ob
22. Ai News (2026-04-07). "Boomi calls it "data activation" and says it's the missing step in every AI deployment." https://www.artificialintelligence-news.com/news/boomi-agentic-ai-data-activation-missing-step/
23. Towards Data Science (2026-04-07). "From 4 Weeks to 45 Minutes: Designing a Document Extraction System for 4,700+ PDFs." https://towardsdatascience.com/from-4-weeks-to-45-minutes-designing-a-document-extraction-system-for-4700-pdfs/
24. Aws Ml (2026-04-06). "Building Intelligent Search with Amazon Bedrock and Amazon OpenSearch for hybrid RAG solutions." https://aws.amazon.com/blogs/machine-learning/building-intelligent-search-with-amazon-bedrock-and-amazon-opensearch-for-hybrid-rag-solutions/
25. Dev To Ai (2026-04-09). "Best AI Chatbot for University Knowledge Bases (2026): CustomGPT.ai vs ChatGPT vs Claude." https://dev.to/alexandria_carlson_21fcf8/best-ai-chatbot-for-university-knowledge-bases-2026-customgptai-vs-chatgpt-vs-claude-4i0m
26. The Register (2026-04-23). "Stale gov.uk pages are feeding AI overviews old data and Brits are believing it." https://go.theregister.com/feed/www.theregister.com/2026/04/23/stale_govuk_pages_are_feeding/
27. Ai News (2026-04-27). "Google warns malicious web pages are poisoning AI agents." https://www.artificialintelligence-news.com/news/google-warns-malicious-web-pages-poisoning-ai-agents/
28. Dev To Ai (2026-04-23). "The EDPB Is Asking About Your AI Agents. Most Teams Can't Answer." https://dev.to/waxell/the-edpb-is-asking-about-your-ai-agents-most-teams-cant-answer-gfk
29. Dev To Ai (2026-04-23). "RAG Pipelines Are the Next Prompt Injection Frontier." https://dev.to/cor_e_fc39e7a5540548c70c0/rag-pipelines-are-the-next-prompt-injection-frontier-kpf
30. Zdnet Ai (2026-04-24). "How indirect prompt injection attacks on AI work - and 6 ways to shut them down." https://www.zdnet.com/article/how-indirect-prompt-injection-attacks-on-ai-work-and-6-ways-to-shut-them-down/
31. MIT Technology Review (2026-04-22). "AI needs a strong data fabric to deliver business value." https://www.technologyreview.com/2026/04/22/1135295/ai-needs-a-strong-data-fabric-to-deliver-business-value/
32. MIT Technology Review (2026-04-27). "Rebuilding the data stack for AI." https://www.technologyreview.com/2026/04/27/1136322/rebuilding-the-data-stack-for-ai/
33. Dev To Ai (2026-04-23). "Add cryptographic authorization to AI agents in 5 minutes." https:

//dev.to/allforscience/add-cryptographic-authorization-to-ai-agents-in-5-minutes-49p1
34. Dev To Ai (2026-04-25). "Building a Cryptographically Secure 4-Agent E-Commerce System: a Practical Guide." https://dev.to/allforscience/building-a-cryptographically-secure-4-agent-e-commerce-system-a-practical-guide-271c
35. Dev To Ai (2026-04-22). "API keys were designed for humans. AI agents break them in 4 ways." https://dev.to/allforscience/api-keys-were-designed-for-humans-ai-agents-break-them-in-4-ways-422
36. Dev To Ai (2026-04-24). "Tian AI vs ChatGPT: Why Local AI Is the Future of Privacy." https://dev.to/3969129510/tian-ai-vs-chatgpt-why-local-ai-is-the-future-of-privacy-b2o
37. Dev To Ai (2026-04-25). "Why Tian AI Runs 100% Offline (And Why That Matters) (v1777123270)." https://dev.to/3969129510/why-tian-ai-runs-100-offline-and-why-that-matters-v1777123270-2kjc
38. Dev To Ai (2026-04-26). "Willow – local-first AI stack, phone reads desktop KB over LAN, no cloud relay." https://dev.to/sean_campbell_840bd62bf7e/willow-local-first-ai-stack-phone-reads-desktop-kb-over-lan-no-cloud-relay-189c

## Legal Services

*capability_id: legal-services · created: 2026-06-05 · revised: 2026-06-05 · refs: 19*

### Near (2026–2031)

For legal services broadly, the delivery mechanism of contract management shifts toward automated initial triage and closing summaries [6][18], while the process of contract drafting integrates new clauses to define the ownership of AI-generated assets, such as prompt libraries [22]. The educational substance of legal advisory remains constrained by a mandatory human-in-the-loop verification layer to mitigate negligence risks [1], as major AI vendors categorize outputs as "for entertainment purposes only" to shift professional liability entirely to the human practitioner [3]. In the European Union, the process of legal & legislative compliance becomes more administratively burdensome due to mandates for Annex IV technical files [4], agentic governance requirements [7], and oversight inquiries from the European Data Protection Board [11], especially for public sector AI use [21]. Within the subsegment of accessibility compliance in the United States, the delivery mechanism of auditing shifts toward AI-powered tools to identify gaps faster than manual audits [20], though the process is slowed by regulatory delays in ADA Title II rules [13][17]. For institutions managing high-stakes campus incidents, the process of dispute resolution faces new liability precedents emerging from criminal investigations into the role of AI in violence [8][14][15]. Final high-stakes authorization is unlikely to change in this horizon because fiduciary duty remains legally tethered to licensed human practitioners [1].

*Hypothesis: In the near term, institutions may reduce time spent on initial contract triage and compliance auditing, but will likely see an increase in operational costs associated with liability verification and regional regulatory adherence.*

### Mid (2036–2041)

For legal services broadly, if current trajectories regarding liability ambiguity and regional regulatory divergence hold, the capability may shift toward the implementation of formalized, machine-readable audit trails to document human oversight [1][2]. Extrapolating from current governance challenges of agentic AI [7] and EU technical documentation requirements [4], institutions could automate the process of legal & legislative compliance through immutable logs of autonomous actions to shield the organization from negligence claims [11]. If the operational cost of maintaining these verification layers—necessary to prevent the submission of AI-generated hallucinations to courts [19]—continues to offset the speed gains of AI, the total cost of institutional legal advisory is likely to remain stagnant [2]. The fundamental requirement for human licensure persists because legal systems resist attributing professional fiduciary duty to non-human entities [1].

*Hypothesis: In the mid term, the capability may evolve into a bifurcated model where AI agents execute high-volume processing and initial triage, while human practitioners focus exclusively on liability-bearing sign-off and risk adjudication.*

### Far (2046–2051)

For legal services broadly, the long-term state of the capability depends on whether legal systems evolve to recognize "bounded autonomy" for AI agents or maintain human exclusivity. One possible outcome involves the convergence of global regulatory regimes, potentially led by the EU's structured approach to technical files [4] and agentic governance [7], which could standardize how institutions manage legal & legislative compliance. A rival outcome involves persistent fragmentation, where divergent regional laws on AI liability—such as legislative attempts to limit lab liability in some US jurisdictions [12] or state-level litigation over AI rules [10]—increase the complexity of institutional legal advisory. It remains unknown if professional indemnity insurance will eventually cover autonomous agentic decisions. The requirement for human-certified legal standing in court resists change even at this horizon due to the social legitimacy requirements of the judiciary.

*Hypothesis A: In the far term, institutional legal services may become a highly standardized compliance function driven by global, machine-readable regulatory frameworks.*

*Hypothesis B (competing): In the far term, institutional legal services may become more costly and fragmented as they navigate a patchwork of conflicting regional liability laws and autonomous agent restrictions.*

## References


1. The Register (2026-04-05). "AI agents promise to 'run the business,' but who is liable if things go wrong?" https://go.theregister.com/feed/www.theregister.com/2026/04/05/ai_agents_liability/
2. Ai Snake Oil (2026-02-12). "AI Won't Automatically Make Legal Services Cheaper" https://www.normaltech.ai/p/ai-wont-automatically-make-legal
3. TechCrunch (2026-04-05). "Copilot is 'for entertainment purposes only,' according to Microsoft's terms of use" https://techcrunch.com/2026/04/05/copilot-is-for-entertainment-purposes-only-according-to-microsofts-terms-of-service/
4. Dev To Ai (2026-04-07). "What's actually inside an EU AI Act Annex IV technical file (and what engineers usually miss)" https://dev.to/rostyslav_myronenko_faf2f/whats-actually-inside-an-eu-ai-act-annex-iv-technical-file-and-what-engineers-usually-miss-4nh4
5. The Register (2026-04-08). "Japan relaxes privacy laws to make itself the 'easiest country to develop AI' " https://go.theregister.com/feed/www.theregister.com/2026/04/08/japan_privacy_law_changes_ai/
6. Dev To Ai (2026-04-08). "Building Guard-Clause: AI-Powered Contract Analysis Without the Legal Team" https://dev.to/jonomor_ecosystem/building-guard-clause-ai-powered-contract-analysis-without-the-legal-team-djj
7. Ai News (2026-04-09). "Agentic AI's governance challenges under the EU AI Act in 2026" https://www.artificialintelligence-news.com/news/agentic-ais-governance-challenges-under-the-eu-ai-act-in-2026/
8. TechCrunch (2026-04-09). "Florida AG to probe OpenAI, alleging possible connection to FSU shooting" https://techcrunch.com/2026/04/09/florida-ag-to-probe-openai-alleging-possible-connection-to-fsu-shooting/
9. Dev To Ai (2026-04-10). "Why Your Source Code Is at Risk When Using AI Coding Assistants" https://dev.to/genevieve_breton_cb795f52/why-your-source-code-is-at-risk-when-using-ai-coding-assistants-29hn
10. Guardian Tech (2026-04-10). "Elon Musk's xAI sues Colorado over new rules for artificial intelligence" https://www.theguardian.com/technology/2026/apr/09/elon-musk-xai-colorado-lawsuit
11. Dev To Ai (2026-04-10). "The EDPB Is Asking About Your AI Agents. Most Teams Can't Answer." https://dev.to/waxell/the-edpb-is-asking-about-your-ai-agents-most-teams-cant-answer-gfk
12. Hacker News (2026-04-10). "OpenAI backs Illinois bill that would limit when AI labs can be held liable" https://www.wired.com/story/openai-backs-bill-exempt-ai-firms-model-harm-lawsuits/
13. Edtech Magazine (2026-04-21). "A Guide to the ADA Title II Accessibility Rule: Deadline Extended to April 2027" https://edtechmagazine.com/higher/article/2025/06/guide-ada-title-ii-accessibility-rule-perfcon
14. BBC News (2026-04-21). "OpenAI faces criminal probe over role of ChatGPT in shooting" https://www.bbc.com/news/articles/c62j4ldp2jqo?at_medium=RSS&at_campaign=rss
15. NPR (2026-04-21). "Florida AG launches criminal investigation into ChatGPT over FSU shooting" https://www.npr.org/2026/04/21/nx-s1-5793967/florida-openai-investigation-mass-shooting-fsu
16. Al Jazeera (2026-04-21). "Pro-Palestine legal aid requests stay high in 2025 amid US campus pressure" https://www.aljazeera.com/news/2026/4/21/pro-palestine-legal-aid-requests-stay-high-in-2025-amid-us-campus-pressure?traffic_source=rss
17. NPR (2026-04-22). "Trump administration delays rule aimed at improving disability access in schools" https://www.npr.org/2026/04/22/nx-s1-5791680/doj-disability-web-access-delay-schools
18. Ai News (2026-04-22). "AI in law firms entering its closing summaries" https://www.artificialintell

igence-news.com/news/ai-in-law-firms-entering-its-closing-summaries/

19. Dev To Ai (2026-04-23). “The Expert’s Exemption” https://dev.to/thesythesis/the-experts-exemption-284i
20. Dev To Ai (2026-04-23). “How Ksolves Is Using AI to Solve Accessibility Compliance Faster Than Traditional Audits Ever Could” https://dev.to/anisha_pal_b1708bf33feb5f/how-ksolves-is-using-ai-to-solve-accessibility-compliance-faster-than-traditional-audits-ever-could-2acm
21. Arxiv Cs Cy (2026-04-28). “Algorithmic Administration and the EU AI Act: Legal Principles for Public Sector Use of AI” https://arxiv.org/abs/2604.22765
22. Dev To Ai (2026-04-28). “Prompt Non-Compete Clauses: When Your Prompt Library Belongs to Your Employer” https://dev.to/velocityai/prompt-non-compete-clauses-when-your-prompt-library-belongs-to-your-employer-3b24

## Library Management

*capability_id: library-management · created: 2026-04-07 · revised: 2026-04-07 · refs: 2*

### Near (2026–2031)

For a subsegment of institutions deploying generative AI assistants for their digital repositories, the mechanism of library collection access management may shift toward hybrid semantic and text-based retrieval [1]. The emergence of AI citation registries suggests that library collection resource management may evolve to include the maintenance of "structured authority" backends to prevent AI inference errors in discovery [2]. This creates a distinction where hybrid RAG architectures alter the delivery mechanism of access [1], while the implementation of authoritative registries alters the educational substance of resource management by redefining how provenance is verified [2]. Processes such as library membership management and the physical management of art/museum collections are unlikely to change in this period due to their dependence on institutional identity systems and physical facility constraints [1][2].

*Hypothesis: In the near term, a subset of institutions may integrate hybrid RAG architectures and structured citation registries to provide more accurate and intent-based access to digital assets.*

### Mid (2036–2041)

For the same subsegment of institutions utilizing AI-driven discovery, if current trajectories in hybrid search and structured authority hold, the process of library collection access management could transition toward personalized, context-aware synthesis of resources [1]. Extrapolating from the requirement for authoritative backends [2], the primary function of the library may shift from managing a searchable catalog to managing the "ground truth" registries that constrain AI inference. However, the core standards for archival preservation will likely persist because the registries themselves depend on high-fidelity, long-term metadata to maintain their authority [2].

*Hypothesis: In the mid term, the delivery of digital collection access for technically advanced libraries may shift from document retrieval to the agentic synthesis of resources verified by structured authority registries.*

### Far (2046–2051)

For institutions operating these agentic discovery layers, the long-term outcome of library collection access management remains uncertain, with two rival outcomes. One scenario involves the total abstraction of the catalog, where the library's primary function is the maintenance of the structured registries that fuel external synthesis agents [1][2]. A competing scenario involves a return to strictly human-curated finding aids if the technical overhead of maintaining "structured authority" fails to prevent systemic hallucinations in AI systems [2]. It remains unknown whether automated registries can replace the nuance of human-led curation in specialized archival contexts. Legal constraints regarding copyright and proprietary database access will likely resist change, limiting the scope of what agentic assistants can retrieve and synthesize.

*Hypothesis A: In the far term, digital library access for the affected subsegment may evolve into a backend-only service providing verified data to autonomous synthesis agents.*

*Hypothesis B (competing): In the far term, a failure of AI registries to ensure epistemic reliability may lead to the restoration of human-curated indices as the primary authority.*

### References


1. Aws Ml (2026-04-06). "Building Intelligent Search with Amazon Bedrock and Amazon OpenSearch for hybrid RAG solutions." https://aws.amazon.com/blogs/machine-learning/building-intelligent-search-with-amazon-bedrock-and-amazon-opensearch-for-hybrid-rag-solutions/
2. Dev To Ai (2026-04-06). "AI Citation Registries as Information Infrastructure for AI Systems." https://dev.to/aigistry/ai-citation-registries-as-information-infrastructure-for-ai-systems-50o6

## Promotions Management

*capability_id: promotions-management · created: 2026-05-28 · revised: 2026-05-28 · refs: 13*

### Near (2026–2031)

For Promotions Management broadly, the process of brand management is shifting toward answer engine optimization (AEO) and generative engine optimization (GEO) to maintain visibility within agent-based discovery systems [3][12]. For a subset of institutions prioritizing algorithmic discovery, the mechanism for visibility involves prioritizing technical accuracy to secure citations in generative AI search results [2][8]. In this subsegment, brand management integrates GEO through the use of specific auditing checklists to monitor institutional presence across large language models [9]. For institutions implementing automated content workflows, campaign management integrates multi-agent pipelines and specialized code skills to produce promotional text and blog posts without human writers [4][5]. For institutions managing digital acquisition, campaign management incorporates a revised audience playbook to account for a decline in general referral traffic and a corresponding increase in high-intent visitors [11]. For a subset of institutions producing digital assets, campaign management incorporates AI-powered video production tools to reduce the cost of creating promotional media [6]. In institutions adopting advanced analytics, the process of market research incorporates AI browser agents to improve the reliability of competitor price tracking and benchmarking by eliminating the need for manual scraper updates [13]. In the same institutions, market research shifts from ad hoc prompting toward repeatable AI workflows to conduct persona-based research [10]. Physical merchandising and event management are unlikely to change yet because they depend on tangible assets and localized human presence. These adjustments represent a change in the delivery mechanism of institutional outreach rather than a change in educational substance.

*Hypothesis: In the near term, promotions management will shift toward agent-based recommendation systems through the adoption of AEO, the use of autonomous content pipelines, and a pivot in campaign management toward high-intent audience acquisition.*

### Mid (2036–2041)

For Promotions Management broadly, if current trajectories toward agent-based discovery hold [3][12], the process of marketing management may evolve into the continuous auditing and correction of synthetic institutional narratives generated by third-party AI agents [1]. Extrapolating from the deployment of autonomous pipelines [4] and emerging technical accuracy standards [2], campaign management could shift from driving traffic to institutional landing pages toward the management of structured data archives that AI agents use for institutional verification [7]. For a subset of institutions, market research may move toward the use of synthetic cohorts where repeatable persona workflows [10] replace traditional focus groups. The underlying educational substance of the institution is likely to persist unchanged, as these developments modify the delivery function of outreach rather than the academic mission.

*Hypothesis: In the mid term, promotions management may shift toward the systemic monitoring and correction of AI-mediated institutional perceptions and the maintenance of agent-readable verification data.*

### Far (2046–2051)

For Promotions Management broadly, outcomes depend on whether discovery systems prioritize optimized technical metadata or externally verifiable institutional outcomes [2]. One plausible outcome is that brand management becomes a bounded technical function of algorithmic alignment with dominant agent architectures [3]. A rival scenario involves a counterproductivity effect where the reliance on autonomous, zero-human content pipelines [4] erodes perceived institutional authenticity, prompting a return to human-curated discovery networks. It remains unknown whether decentralized verification protocols will supersede centralized AI search gateways. Physical event management resists change even at this horizon due to the persistent social legitimacy requirement of in-person academic gathering.

*Hypothesis A: In the far term, promotions management becomes a specialized technical operation focused* *on the algorithmic alignment of institutional data with agent discovery protocols.*

*Hypothesis B (competing): In the far term, a crisis of authenticity caused by synthetic content leads to a restoration of human-curated brand management and high-touch event-based promotion.*

## References


1. The Verge (2026-04-06). "Can AI responses be influenced? The SEO industry is trying." https://www.theverge.com/tech/900302/ai-seo-industry-google-search-chatgpt-gemini-marketing
2. Dev To Ai (2026-04-06). "The 2026 AI Governance Framework: Why Technical Accuracy is the New SEO Gold Standard." https://dev.to/mirakade/the-2026-ai-governance-framework-why-technical-accuracy-is-the-new-seo-gold-standard-1fn6
3. Dev To Ai (2026-04-06). "What is AEO, and How to Do AEO by Agent." https://dev.to/_bdc022b88aec19efbe2f98/what-is-aeo-and-how-to-do-aeo-by-agent-4l9k
4. Dev To Ai (2026-04-06). "Inside Our Autonomous AI Pipeline: 4 Agents, Zero Human Writers." https://dev.to/sebastian_chedal/inside-our-autonomous-ai-pipeline-4-agents-zero-human-writers-md8
5. Dev To Ai (2026-04-07). "I Built a Claude Code Skill That Writes SEO-Optimized Blog Posts From Scratch." https://dev.to/whoffagents/i-built-a-claude-code-skill-that-writes-seo-optimized-blog-posts-from-scratch-2651
6. Google AI Blog (2026-04-02). "Create, edit and share videos at no cost in Google Vids." https://blog.google/products-and-platforms/products/workspace/google-vids-updates-lyria-veo/
7. Dev To Ai (2026-04-10). "Breaking the Ice: How to Get Your Brand Noticed by AI Search Engines for the First Time." https://dev.to/team_hyperpitch/breaking-the-ice-how-to-get-your-brand-noticed-by-ai-search-engines-for-the-first-time-51fh
8. Dev To Ai (2026-04-10). "How to Get Cited by AI Search Engines: ChatGPT, Perplexity & Claude in 2026." https://dev.to/iris1031/how-to-get-cited-by-ai-search-engines-chatgpt-perplexity-claude-in-2026-21p0
9. Dev To Ai (2026-04-21). "Checklist GEO 30 points : audit complet de ton site 2026." https://dev.to/augustin_foucheres_828af4/checklist-geo-30-points-audit-complet-de-ton-site-2026-4od4
10. Towards Data Science (2026-04-22). "From Ad Hoc Prompting to Repeatable AI Workflows with Claude Code Skills." https://towardsdatascience.com/from-ad-hoc-prompting-to-repeatable-ai-workflows-with-claude-code-skills/
11. Fast Company (2026-04-24). "AI search demands a new audience playbook." https://www.fastcompany.com/91529130/ai-search-demands-a-new-audience-playbook?partner=rss&utm_source=rss&utm_medium=feed&utm_campaign=rss+fastcompany&utm_content=rss
12. Dev To Ai (2026-04-25). "Your Website Works. Your SEO Is Fine. So Why Are You Still Invisible?" https://dev.to/andrei_mironiuk_1d2c01e06/your-website-works-your-seo-is-fine-so-why-are-you-still-invisible-d3f
13. Dev To Ai (2026-04-27). "AI Browser Agents in 2026: Stagehand vs Browser Use vs Playwright." https://dev.to/alexcloudstar/ai-browser-agents-in-2026-stagehand-vs-browser-use-vs-playwright-38ob

## Research Administration

*capability_id: research-administration · created: 2026-05-17 · revised: 2026-05-17 · refs: 9*

### Near (2026–2031)

For research infrastructure management broadly, the process of resource allocation is incorporating transparent screening frameworks to estimate the inference and training impacts of LLM projects [1], while the deployment of specialized hardware allows for full precision training of models with over 100B parameters on single GPUs [2]. Institutions are navigating enterprise GPU shortages and constraints in power and management controller silicon by employing observability stacks to monitor hardware efficiency [3][4][5], even as infrastructure from NVIDIA and Google reduces specific inference costs [6]. In the United States, research compliance management is aligning with a National AI Legislative Framework [7], including stricter vetting of foreign researchers [8] and new export control protocols to prevent the exploitation of U.S. AI models by restricted foreign firms [9][10][11]. For the healthcare and life sciences subsegment, the delivery mechanism of research compliance management is integrating human-in-the-loop agentic workflows [12] to address risks from autonomous lab experiments that can design and run thousands of trials without human intervention [13][14]. Core institutional review board (IRB) processes are unlikely to change in this period due to the high social legitimacy requirements for human adjudication of ethical risk.

*Hypothesis: In the near term, research infrastructure management will move toward transparent resource screening and diversified local hardware, while US-based and life-science compliance processes will integrate federal mandates and agentic automation for biosafety and export controls.*

### Mid (2036–2041)

For a subset of institutions utilizing high-performance compute, if current trajectories in energy efficiency hold, a reduction in energy use by up to 100x may decrease the operational costs of maintaining local AI resources, potentially reducing institutional dependency on third-party cloud providers [15]. In regions prioritizing “frugal AI” strategies, research infrastructure management may shift toward the procurement and maintenance of smaller, specialized model architectures rather than general-purpose frontier systems [16]. For the healthcare and life sciences subsegment, if agentic workflows for regulatory filings scale, research compliance management may shift from the manual preparation of documentation to the systemic auditing of agent-generated submissions [12]. This represents a change in the educational substance of compliance, moving from clerical accuracy to the validation of algorithmic provenance. The fundamental legal accountability of the Principal Investigator (PI) for research misconduct is unlikely to change, as institutional liability remains tethered to human legal personhood.

*Hypothesis: In the mid term, research infrastructure management for high-compute institutions may shift toward energy-efficient local autonomy, while compliance in specialized subsegments will move from document generation to the auditing of autonomous agentic workflows.*

### Far (2046–2051)

For institutions managing frontier AI infrastructure, the capability may diverge based on the evolution of compute physics. One outcome involves the integration of quantum technology that disrupts current AI training and inference tasks, potentially rendering existing GPU-based infrastructure management processes obsolete [17]. Alternatively, if quantum utility remains limited to specific niches, infrastructure management may stabilize around highly efficient, bounded-autonomy GPU clusters. The precise timing and scale of this transition remain unknown due to uncertainty in quantum hardware scaling. Fundamental regulatory requirements for the oversight of dual-use biological and digital technologies will likely resist change even at this horizon.

*Hypothesis A: In the far term, research infrastructure management shifts toward quantum-native architectures, rendering legacy GPU-based resource allocation processes obsolete.*

*Hypothesis B (competing): In the far term, infrastructure management stabilizes around refined, high-*

*efficiency classical compute clusters as quantum technology fails to scale for general AI workloads.*

## References


1. Arxiv Cs Lg (2026-04-23). "Transparent Screening for LLM Inference and Training Impacts." https://arxiv.org/abs/2604.19757
2. Hacker News (2026-04-08). "MegaTrain: Full Precision Training of 100B+ Parameter LLMs on a Single GPU." https://arxiv.org/abs/2604.05091
3. Dev To Ai (2026-04-10). "How To Navigate Enterprise GPU Shortages for AI Workloads." https://dev.to/autonainews/how-to-navigate-enterprise-gpu-shortages-for-ai-workloads-4dkj
4. The Register (2026-04-23). "AI now gobbling up power and management chips for servers." https://go.theregister.com/feed/www.theregister.com/2026/04/23/ai_now_gobbling_up_power/
5. The Register (2026-04-23). "Datadog digs down into GPU efficiency as AI costs soar." https://go.theregister.com/feed/www.theregister.com/2026/04/23/datadog_digs_down_into_gpu/
6. Ai News (2026-04-23). "NVIDIA and Google infrastructure cuts AI inference costs." https://www.artificialintelligence-news.com/news/nvidia-and-google-infrastructure-cuts-ai-inference-costs/
7. EDUCAUSE Review (2026-04-21). "White House Unveils National AI Legislative Framework." https://er.educause.edu/articles/2026/4/white-house-unveils-national-ai-legislative-framework
8. Scmp Tech (2026-04-22). "US visa system 'rolls out red carpet' for Chinese spies, congressional hearing told." https://www.scmp.com/news/china/diplomacy/article/3351062/us-visa-system-rolls-out-red-carpet-chinese-spies-congressional-hearing-told?utm_source=rss_feed
9. BBC News (2026-04-23). "White House memo claims mass AI theft by Chinese firms." https://www.bbc.com/news/articles/cpqxgxx9nrqo?at_medium=RSS&at_campaign=rss
10. NPR (2026-04-24). "Trump administration vows crackdown on Chinese firms 'exploiting' U.S. AI models." https://www.npr.org/2026/04/24/g-s1-118582/administration-crackdown-on-chinese-firms-exploiting-u-s-ai-models
11. Fast Company (2026-04-24). "Trump administration vows crackdown on China's 'exploiting' of AI models made in the U.S." https://www.fastcompany.com/91532063/trump-administration-vows-crackdown-chinas-exploiting-ai-models-made-u-s?partner=rss&utm_source=rss&utm_medium=feed&utm_campaign=rss+fastcompany&utm_content=rss
12. Aws Ml (2026-04-08). "Human-in-the-loop constructs for agentic workflows in healthcare and life sciences." https://aws.amazon.com/blogs/machine-learning/human-in-the-loop-constructs-for-agentic-workflows-in-healthcare-and-life-sciences/
13. The Conversation Tech (2026-04-09). "AI can design and run thousands of lab experiments without human hands. Humanity isn't ready for the new risks this brings to biology." https://theconversation.com/ai-can-design-and-run-thousands-of-lab-experiments-without-human-hands-humanity-isnt-ready-for-the-new-risks-this-brings-to-biology-279191
14. Fast Company (2026-04-13). "AI is rewriting the rules of biological experiments, but safety regulations aren't keeping up." https://www.fastcompany.com/91524571/ai-biology-lab-experiments-risks?partner=rss&utm_source=rss&utm_medium=feed&utm_campaign=rss+fastcompany&utm_content=rss
15. Science Daily Comp (2026-04-05). "AI breakthrough cuts energy use by 100x while boosting accuracy." https://www.sciencedaily.com/releases/2026/04/260405003952.htm
16. Rest Of World (2026-04-02). "Nations priced out of Big AI are building with frugal models." https://restofworld.org/2026/frugal-ai-big-tech/?utm_source=rss&utm_medium=rss&utm_campaign=feeds
17. Scmp Tech (2026-04-14). "Chinese team shows quantum tech can disrupt AI in a real world task." https://www.scmp.com/news/china/science/article/3349995/chinese-team-shows-quantum-tech-can-disrupt-ai-real-world-task?utm_source=rss_feed

## Research Delivery

*capability_id: research-delivery · created: 2026-06-03 · revised: 2026-06-03 · refs: 27*

### Near (2026–2031)

For a subset of institutions focusing on computational, biomedical, mathematical, and astronomical research, and in specific regions including Japan, the UK, and South Korea, AI is altering both the delivery mechanisms and the educational substance of research production, dataset management, and output management. In research production for the life sciences, delivery is shifting toward autonomous execution via agentic discovery, where AI designs and runs lab experiments without human intervention [19], accelerates drug discovery through petabyte-scale data scanning [8], and orchestrates multi-step workflows for drug molecule evaluation and optimization through hierarchical agent skills [41]. This production is supported by specialized bioinformatics skill-sets [21], evidence-based medical discovery agents [32], and standardized benchmarks like LABBench2 [28]. Technical research production is accelerated in software-heavy domains by agentic models capable of executing full-stack coding [38][39] and the automation of GPU kernel optimization [4], yet remains fundamentally limited in experimental physics, where agents have shown an inability to end-to-end reproduce numerical results from experimental papers [43]. Research output management delivery is shifting toward automated manuscript synthesis via multi-agent frameworks [18][20], AI-integrated co-authoring tools [36], the automation of scientific figure generation [40], and custom extraction pipelines for literature reviews [7][23]. Research dataset management is diverging regionally—where Japan's relaxed privacy laws accelerate data availability [10] while the UK faces systemic hurdles in centralizing public data [11]—and technically, through the use of privacy-preserving federated learning [24][25], the generation of controllable synthetic datasets [30], and end-to-end lineage tools [31]. However, research production in behavioral and cognitive sciences faces a substance risk where LLM behavioral simulators lack causal validity [3], a failure mode compounded by "cognitive surrender" where researchers abandon logical verification in favor of AI-generated results [1]. The institutional peer-review process for research output management is unlikely to change yet, as it remains the primary mechanism for professional accountability and social legitimacy.

*Hypothesis: In the near term, research delivery within AI-reliant subsegments will experience a tension between accelerated technical output and a decline in human-led logical and causal verification.*

### Mid (2036–2041)

For these same subsegments and regions, if current trajectories hold, research production may shift toward a verification-centric model. Because "cognitive surrender" [1] and causally invalid simulations [3] erode the validity of automated results, the substance of research production may shift from primary discovery to the rigorous verification of AI-generated hypotheses. The integration of documented human-in-the-loop validation steps, particularly in healthcare and life sciences [16], could become a requirement within research output management to mitigate biological risks [19][27] and maintain scientific legitimacy. In technical research production, the emergence of AI-to-AI communication protocols [14] may allow autonomous agents to manage the iterative loop of hypothesis testing and refinement independently, provided the "physics gap" in numerical reproduction [43] is bridged. Despite these shifts, the fundamental requirement for human authors to assume legal and ethical responsibility for research outputs will likely persist due to existing institutional governance and liability frameworks.

*Hypothesis: In the mid term, the core substance of research production in these fields may transition from the act of discovery to the act of auditing and validating agent-generated claims.*

### Far (2046–2051)

For these same subsegments and regions, the long-term trajectory depends on whether the volume of AI-generated output preserves or destroys the signal-to-noise ratio of scientific literature. One outcome involves a complete decoupling of "production" from "authorship," where the delivery of research production is fully agentic and human roles are limited to high-level goal setting and ethical oversight. A rival outcome suggests a "retrenchment to slow science," where institutions reject automated research output management in favor of strictly human-verified proofs to combat the proliferation of plausible but false AI-generated findings. It remains unknown whether AI can move beyond the role of a simulator to achieve genuine causal reasoning in the behavioral and physical sciences. Even at this horizon, the social requirement for a human "principal investigator" to vouch for the integrity of a dataset is likely to resist change due to the nature of professional trust and institutional liability.

*Hypothesis A: In the far term, research delivery in these subsegments evolves into a curated ecosystem of autonomous agentic loops overseen by human auditors.*

*Hypothesis B (competing): In the far term, the collapse of trust in automated research production leads to a systemic return to human-centric, slow-verification protocols for scientific legitimacy.*

## References


1. Ars Technica (2026-04-03). ""Cognitive surrender" leads AI users to abandon logical thinking, research finds." https://arstechnica.com/ai/2026/04/research-finds-ai-users-scarily-willing-to-surrender-their-cognition-to-llms/
2. Arxiv Cs Cy (2026-04-06). "When simulations look right but causal effects go wrong: Large language models as behavioral simulators." https://arxiv.org/abs/2604.02458
3. Marktechpost (2026-04-06). "RightNow AI Releases AutoKernel: An Open-Source Framework that Applies an Autonomous Agent Loop to GPU Kernel Optimization for Arbitrary PyTorch Models." https://www.marktechpost.com/2026/04/06/rightnow-ai-releases-autokernel-an-open-source-framework-that-applies-an-autonomous-agent-loop-to-gpu-kernel-optimization-for-arbitrary-pytorch-models/
4. Dev To Ai (2026-04-07). "Automate Your Literature Review: Build a Custom AI Extraction Pipeline." https://dev.to/ken_deng_ai/automate-your-literature-review-build-a-custom-ai-extraction-pipeline-1dmd
5. The Conversation Tech (2026-04-07). "AI is reengineering drug discovery by speeding up testing and scanning petabytes of data for connections between diseases." https://theconversation.com/ai-is-reengineering-drug-discovery-by-speeding-up-testing-and-scanning-petabytes-of-data-for-connections-between-diseases-274693
6. The Register (2026-04-08). "Japan relaxes privacy laws to make itself the 'easiest country to develop AI'." https://go.theregister.com/feed/www.theregister.com/2026/04/08/japan_privacy_law_changes_ai/
7. The Register (2026-04-08). "UK's grand plan to fuel AI with public data faces uphill battle." https://go.theregister.com/feed/www.theregister.com/2026/04/08/national_data_library_plan/
8. The Register (2026-04-08). "Talk ain't cheap: DARPA offers grants for new AI-to-AI communication protocol." https://go.theregister.com/feed/www.theregister.com/2026/04/08/darpa_wants_ai_agent_communication/
9. Aws Ml (2026-04-08). "Human-in-the-loop constructs for agentic workflows in healthcare and life sciences." https://aws.amazon.com/blogs/machine-learning/human-in-the-loop-constructs-for-agentic-workflows-in-healthcare-and-life-sciences/
10. Marktechpost (2026-04-09). "Google AI Research Introduces PaperOrchestra: A Multi-Agent Framework for Automated AI Research Paper Writing." https://www.marktechpost.com/2026/04/08/google-ai-research-introduces-paperorchestra-a-multi-agent-framework-for-automated-ai-research-paper-writing/
11. The Conversation Tech (2026-04-09). "AI can design and run thousands of lab experiments without human hands. Humanity isn't ready for the new risks this brings to biology." https://theconversation.com/ai-can-design-and-run-thousands-of-lab-experiments-without-human-hands-humanity-isnt-ready-for-the-new-risks-this-brings-to-biology-279191
12. Dev To Ai (2026-04-09). "Building a Multi-Agent Research System with LangGraph: How I Taught Three AI Agents to Collaborate." https://dev.to/benturtle/building-a-multi-agent-research-

system-with-langgraph-how-i-taught-three-ai-agents-to-collaborate-2ng6
13. Dev To Ai (2026-04-10). “Add 197 Bioinformatics Skills to Claude Code with SciAgent-Skills.” https://dev.to/gentic_news/add-197-bioinformatics-skills-to-claude-code-with-sciagent-skills-32gh
14. OpenAI (2026-04-10). “Research with ChatGPT.” https://openai.com/academy/search-and-deep-research
15. Dev To Ai (2026-04-11). “Federated Learning in 2026: Privacy-Preserving AI.” https://dev.to/lufumeiying/federated-learning-in-2026-privacy-preserving-ai-5f9j
16. Dev To Ai (2026-04-11). “Why N=1 Rare Disease Sites Are Excluded from Federated Learning by Architecture — and What Outcome Routing Does Instead.” https://dev.to/roryqis/why-n1-rare-disease-sites-are-excluded-from-federated-learning-by-architecture-and-what-outcome-b9c
17. Fast Company (2026-04-13). “AI is rewriting the rules of biological experiments, but safety regulations aren’t keeping up.” https://www.fastcompany.com/91524571/ai-biology-lab-experiments-risks?partner=rss&utm_source=rss&utm_medium=feed&utm_campaign=rss+fastcompany&utm_content=rss
18. Arxiv Cs Ai (2026-04-14). “LABBench2: An Improved Benchmark for AI Systems Performing Biology Research.” https://arxiv.org/abs/2604.09554
19. Marktechpost (2026-04-21). “Google Introduces Simula: A Reasoning-First Framework for Generating Controllable, Scalable Synthetic Datasets Across Specialized AI Domains.” https://www.marktechpost.com/2026/04/21/google-introduces-simula-a-reasoning-first-framework-for-generating-controllable-scalable-synthetic-datasets-across-specialized-ai-domains/
20. Aws Ml (2026-04-21). “End-to-end lineage with DVC and Amazon SageMaker AI MLflow apps.” https://aws.amazon.com/blogs/machine-learning/end-to-end-lineage-with-dvc-and-amazon-sagemaker-ai-mlflow-apps/
21. Arxiv Cs Ai (2026-04-21). “DeepER-Med: Advancing Deep Evidence-Based Research in Medicine Through Agentic AI.” https://arxiv.org/abs/2604.15456
22. The Register (2026-04-23). “Microsoft gives your Word documents an AI co-author you didn’t ask for.” https://go.theregister.com/feed/www.theregister.com/2026/04/23/microsoft_gives_your_word_documents/
23. Marktechpost (2026-04-23). “OpenAI Releases GPT-5.5, a Fully Retrained Agentic Model That Scores 82.7% on Terminal-Bench 2.0 and 84.9% on GDPval.” https://www.marktechpost.com/2026/04/23/openai-releases-gpt-5-5-a-fully-retrained-agentic-model-that-scores-82-7-on-terminal-bench-2-0-and-84-9-on-gdpval/
24. Simon Willison (2026-04-25). “Quoting Romain Huet.” https://simonwillison.net/2026/Apr/25/romain-huet/#atom-everything
25. Dev To Ai (2026-04-27). “How AI Figure Generation Is Changing Scientific Visualization in 2026.” https://dev.to/ethel_pycha_0fbab9df84f86/how-ai-figure-generation-is-changing-scientific-visualization-in-2026-27h9
26. Arxiv Cs Ai (2026-04-27). “MolClaw: An Autonomous Agent with Hierarchical Skills for Drug Molecule Evaluation, Screening, and Optimization.” https://arxiv.org/abs/2604.21937
27. Lesswrong (2026-04-27). “AI Is Bad at Physics.” https://www.lesswrong.com/posts/c4Kopt4bY3ZbMdshJ/ai-is-bad-at-physics

## Research Funding

*capability_id: research-funding · created: 2026-05-19 · revised: 2026-05-19 · refs: 3*

### Near (2026–2031)

For region-specific government contexts and specialized grant-seeking subsegments, AI is being applied to the identification of new leads and the management of existing funds. In certain government settings, the process of research funds management may involve automated thematic filtering where large language models identify grants for elimination based on ideological or thematic markers [1]. This represents a change in educational substance, as the criteria for funding validity shift from expert peer review toward algorithmic alignment. For a subsegment of grant seekers, the process of research funding identification is shifting from manual directory searches to AI-automated prospecting and personalized lead generation [2]. Additionally, for niche subsegments focusing on transformative AI and animal welfare, new rapid-disbursement funding streams are emerging that operate on accelerated timelines [3]. The legal mechanisms for the actual disbursement of funds are unlikely to change yet due to the rigidity of statutory appropriation laws and government financial regulations.

*Hypothesis: In the near term, region-specific government research funds management may incorporate automated filtering for grant elimination, while specialized subsegments adopt AI-driven automation for research funding identification.*

### Mid (2036–2041)

For region-specific government contexts and specialized grant-seeking subsegments, if current trajectories hold, the interaction between automated identification and automated filtering may create a feedback loop of prompt optimization. Extrapolating from current use, researchers seeking government funds may treat research funding identification as a prompt-engineering exercise, tailoring proposal language to avoid the algorithmic triggers used in research funds management [1]. In the grant-seeking subsegment, the use of automated prospecting may lead to a saturation of high-probability leads, potentially increasing the volume of personalized but low-substance outreach to foundations [2]. Despite these shifts, the administrative reporting requirements for how funds are spent will likely persist unchanged because they serve as the primary mechanism for legal and financial accountability.

*Hypothesis: In the mid term, research funding identification may shift toward a prompt-optimization model where proposals are engineered to bypass algorithmic filters used in fund management.*

### Far (2046–2051)

For region-specific government contexts and specialized grant-seeking subsegments, the long-term interaction between automated prospecting and algorithmic elimination remains uncertain. One possibility is that these tools evolve into sophisticated systems that optimize grant portfolios based on complex data patterns, although it is unknown if these patterns correlate with genuine scientific value. A rival outcome involves the systemic erosion of the social legitimacy of government grants if the management process is perceived as purely ideological rather than meritocratic. The fundamental requirement for financial auditing and the prevention of fraud will likely resist change even at this horizon, as these functions are tied to legal liability rather than thematic preference.

*Hypothesis A: In the far term, government research funding management may transition to a bounded autonomous system that optimizes grant portfolios based on predefined state objectives.*

*Hypothesis B (competing): In the far term, the perceived lack of meritocratic rigor in algorithmic filtering may lead to a migration of research talent toward non-governmental funding sources.*

### References


1. Fast Company (2026-04-14). "What good AI in government actually looks like." https://www.fastcompany.com/91525531/grantwell-what-good-ai-in-government-actually-looks-like?partner=rss&u

tm_source=rss&utm_medium=feed&utm_campaign=rss+fastcompany&utm_content=rss
2. Dev To Ai (2026-04-22). "From Scattershot to Sniper: AI for Smarter Grant Lead Generation." https://dev.to/ken_deng_ai/from-scattershot-to-sniper-ai-for-smarter-grant-lead-generation-4if4
3. Lesswrong (2026-04-24). "Manifund's Falcon Fund." https://www.lesswrong.com/posts/nduaR8YP9omEKuzNJ/manifund-s-falcon-fund

## Research Impact

*capability_id: research-impact · created: 2026-04-11 · revised: 2026-04-11 · refs: 2*

### Near (2026–2031)

For researchers managing non-commercial research impact via digital dissemination, the process of visibility management may shift from search engine optimization (SEO) toward generative engine optimization (GEO) [1]. Because users increasingly employ LLMs as a discovery layer that synthesizes answers from a retrieval set, the mechanism for achieving impact shifts from winning a click to becoming a verifiable and extractable source for model synthesis [1]. Evidence suggests that legacy web authority metrics, such as Domain Authority (DA), no longer reliably predict whether AI platforms will cite a specific source [2]. This decoupling of domain prestige from AI citability suggests that non-commercial research impact management must prioritize content structure over traditional domain authority [2]. This represents a change in the delivery mechanism of the capability, as researchers may alter how they structure digital content to ensure it is "RAG-friendly" [1]. However, core metrics for research impact, such as h-index or formal citations in peer-reviewed journals, are unlikely to change in this period due to their deep embedding in institutional tenure and funding regulations.

*Hypothesis: In the near term, researchers utilizing digital dissemination may adapt their visibility strategies to prioritize machine-readability and structural verifiability over traditional domain authority to increase their presence within generative answer engines.*

### Mid (2036–2041)

For the same subset of researchers focused on web-based dissemination, if current trajectories hold, the process of non-commercial research impact management could transition toward formalized GEO standards. Extrapolating from the requirement for verifiable sources in RAG-based systems [1], researchers may adopt specific metadata schemas to ensure their work is correctly attributed during LLM synthesis. This could shift the educational substance of "impact" from simple reach (clicks or views) to "synthesized utility," where the value of research is measured by its role as a foundational component of AI-generated answers. Despite this, the process of commercial research outcomes—such as patent filing and licensing—will likely persist unchanged because these rely on legal protections and proprietary disclosures rather than public discovery layers.

*Hypothesis: In the mid term, non-commercial research impact for web-active researchers may be increasingly evaluated based on the frequency and accuracy of their work's synthesis within generative engines.*

### Far (2046–2051)

For researchers using digital discovery layers, the capability of managing non-commercial research impact remains subject to the tension between algorithmic visibility and institutional trust. One plausible outcome is that the retrieval set becomes the primary arbiter of academic influence, effectively automating the discovery process for a large portion of non-commercial impact. Conversely, a rival outcome may emerge if the over-optimization of research for AI synthesis leads to a counterproductive erosion of trust, where the "gaming" of GEO undermines the perceived validity of the research itself. It remains unknown whether funding bodies will eventually replace traditional citation counts with generative synthesis metrics. The requirement for human-peer-validated novelty will likely resist change even at this horizon, as it provides the social legitimacy that algorithmic synthesis cannot generate.

*Hypothesis A: In the far term, digital research impact may be primarily mediated through the precision and frequency with which a scholar's work is retrieved and synthesized by autonomous agents.*

*Hypothesis B (competing): In the far term, institutional distrust in AI-mediated visibility may lead to a re-valuation of gated, non-algorithmic verification processes to ensure research impact is not artificially inflated via GEO.*

## References


1. Dzone Ai (2026-04-06). "Engineering for the Answer Engine: GEO for RAG-Friendly Web Apps (TalentHacked.com Case Study)." https://feeds.dzone.com/link/23558/17314715/engineering-for-the-answer-engine-geo-for-rag-frie
2. Dev To Ai (2026-04-10). "Why DA Is Irrelevant for AI Citations (Data from 7 Site Audits)." https://dev.to/chudi_nnorukam/why-da-is-irrelevant-for-ai-citations-data-from-7-site-audits-1il0

## Research Improvement

*capability_id: research-improvement · created: 2026-04-23 · revised: 2026-04-23 · refs: 3*

### Near (2026–2031)

For a subset of research quality management involving biology research, behavioral simulations, and the scientific drafting process, the introduction of specialized benchmarks and automated feedback mechanisms is altering validation protocols. In biology research, the deployment of standardized benchmarks such as LABBench2 enables more precise evaluation of AI systems, shifting the delivery mechanism of quality management toward automated performance auditing [2]. In behavioral simulation, a disconnect between the visual plausibility of LLM outputs and their actual causal accuracy necessitates new validation criteria that prioritize causal verification over output coherence [1]. This represents a change in research substance, as "valid evidence" in simulations must shift from plausibility to verified causal effects [1]. Additionally, the use of LLMs to generate constructive feedback on scientific papers based on author responses is altering the research quality management process during the drafting and revision phase [3]. Researcher performance management is unlikely to change in this horizon, as institutional metrics for productivity typically rely on output volume rather than the technical fidelity of AI-assisted methodologies.

*Hypothesis: In the near term, research quality management for specific AI-driven subsegments may shift toward requiring standardized benchmark validation, explicit causal verification, and AI-mediated constructive feedback.*

### Mid (2036–2041)

For the same subsegments of biology, behavioral, and drafting processes, if current trajectories toward standardized benchmarking and causal auditing hold, research quality management may integrate these technical audits as mandatory prerequisites for publication [1][2]. Extrapolating from the use of tools like LABBench2 and GoodPoint, the process of research quality management could transition from human-led peer review of final results to a technical verification of the AI's operational logic, causal architecture, and feedback iterations [1][2][3]. This shift moves the point of intervention in quality management from the end of the research cycle to the tool-selection and execution phases. The reliance on institutional review boards for final methodological approval is likely to persist, as these bodies provide the social legitimacy required for high-stakes scientific claims.

*Hypothesis: In the mid term, the validation of AI-driven research in these subsegments may evolve into a formal technical auditing process that separates surface-level plausibility from structural validity.*

### Far (2046–2051)

For researchers utilizing synthetic behavioral agents and AI-driven biological discovery, the stability of research quality management depends on whether the gap between plausibility and causal accuracy is fundamentally resolved in model architectures [1]. One plausible outcome involves a specialized regime of "synthetic provenance," where AI-generated data is strictly tiered by its verified causal reliability according to evolving benchmarks [2]. A rival outcome suggests a systemic devaluation of LLM-simulated data for high-stakes causal claims if persistent inaccuracies remain, leading to a renewed institutional preference for empirical human data. It remains unknown whether future architectures can inherently solve the plausibility-accuracy gap or if external, independent verification will always be a requirement. The fundamental requirement for empirical grounding in behavioral and biological sciences will resist change even at this horizon.

*Hypothesis A: In the far term, research quality management may successfully incorporate a tiered system of synthetic data reliability based on rigorous, automated causal verification.*

*Hypothesis B (competing): In the far term, a persistent gap between plausibility and causal accuracy may lead to the systemic devaluation of AI-simulated data in favor of empirical human data.*

## References


1. Arxiv Cs Cy (2026-04-06). "When simulations look right but causal effects go wrong: Large language models as behavioral simulators." https://arxiv.org/abs/2604.02458
2. Arxiv Cs Ai (2026-04-14). "LABBench2: An Improved Benchmark for AI Systems Performing Biology Research." https://arxiv.org/abs/2604.09554
3. Arxiv Cs Ai (2026-04-16). "GoodPoint: Learning Constructive Scientific Paper Feedback from Author Responses." https://arxiv.org/abs/2604.11924

## Research Opportunity Analysis

*capability_id: research-opportunity-analysis · created: 2026-05-28 · revised: 2026-05-28 · refs: 2*

### Near (2026–2031)

For research-intensive environments broadly, the process of research prioritisation may shift toward automated gap detection through the use of parallel AI research agents to map knowledge voids across scientific corpora [1]. This represents a change in the delivery mechanism of research prioritisation, where agentic scanning replaces some manual literature synthesis. Simultaneously, for institutions in regions implementing national-level frontier AI partnerships, such as the Republic of Korea, collaboration opportunity management and research programme development may be accelerated by integrating frontier AI models into strategic scientific planning [2]. Because these tools identify informational voids rather than strategic or social value, the final selection of research directions will likely remain a human-led activity. The social legitimacy of a research agenda depends on expert human validation, making a fully automated prioritisation process unlikely in this horizon.

*Hypothesis: AI agents will support the initial stage of knowledge gap identification and the acceleration of strategic collaboration in the near term.*

### Mid (2036–2041)

For institutions implementing agentic workflows in their research offices, if current trajectories in automated mapping hold, these tools could alter the substance of research programme development. Extrapolating from current demonstrations of parallel agentic gap analysis [1], the identification of “white space” in scientific literature may become a standardized input for designing multi-year research programmes. This shift may move programme development away from intuition-based hypothesis generation toward a model focused on data-driven novelty detection. However, the requirement for peer-reviewed validation of a research gap’s significance will likely persist, as the consensus required for academic legitimacy remains a human social process.

*Hypothesis: Research prioritisation may evolve into a hybrid model where AI agents propose potential knowledge gaps for human curation in the mid term.*

### Far (2046–2051)

For research-intensive environments, the long-term impact on research opportunity analysis remains uncertain, specifically regarding whether AI can identify conceptual paradigm shifts or merely informational gaps. One possibility is that automated gap mapping becomes the primary driver of research agendas, potentially leading to a counterproductive cycle where researchers pursue a high volume of superficially novel but low-impact “gaps” identified by AI. Conversely, a reaction against this automation may occur, where the value of theory-driven rather than gap-driven research is re-emphasized to maintain intellectual substance. The physical constraint of empirical validation—the need to conduct actual experiments or observations—will resist automation even at this horizon.

*Hypothesis A: AI-driven gap analysis becomes the primary mechanism for research prioritisation in the far term.*

*Hypothesis B (competing): The proliferation of AI-identified gaps leads to a crisis of significance, triggering a return to human-centric, theory-led research prioritisation in the far term.*

### References


1. Dev To Ai (2026-04-10). “We mapped 153 gaps in science using 5 parallel AI research agents.” https://dev.to/minimal-architect/we-mapped-153-gaps-in-science-using-5-parallel-ai-research-agents-5fdk
2. Google DeepMind (2026-04-27). “Announcing our partnership with the Republic of Korea.” https://deepmind.google/blog/announcing-our-partnership-with-the-republic-of-korea/

## Research Publications

*capability_id: research-publications · created: 2026-04-11 · revised: 2026-04-11 · refs: 7*

### Near (2026–2031)

For research publications broadly, research publication management is increasingly defined by a tension between automated drafting and the deployment of provenance-tracking infrastructure. Publishers are employing AI-detection tools to maintain the integrity of submitted research outputs [1], while the emergence of AI citation registries suggests a shift toward using structured information infrastructure to verify the authenticity of references [2]. For a subsegment of researchers, the delivery mechanism of research publication management is shifting toward the use of multi-agent frameworks, such as PaperOrchestra, to automate the drafting of manuscripts [5]. Regarding research output reporting, a subsegment of researchers is adapting output formatting—utilizing direct Q&A and tables—to increase visibility within AI search engines [6]. This shift is supported by the development of AI visibility infrastructure designed to interface directly with AI systems [4] and evidence that AI citations are driven by specific "citability" factors rather than traditional domain authority (DA) [7]. In institution-specific settings, such as the Alignment Journal, experimental policies are introducing reviewer compensation to modify the peer-review workflow [3]. The social legitimacy of the human-led peer-review process is unlikely to change in this window because institutional tenure and promotion regimes rely on human intellectual accountability.

*Hypothesis: In the near term, research publication management will be characterized by a technical arms race between multi-agent drafting tools and the infrastructure required to detect and verify AI-generated content.*

### Mid (2036–2041)

For research publications broadly, if current trajectories in registry adoption and detection instability hold, the mechanism of verification may shift from reactive detection to proactive provenance. Extrapolating from the limitations of binary AI-text detection [1], publishers may require research output reporting to be anchored in verified citation registries [2] to prevent the proliferation of fabricated evidence. Research output reporting may further diverge from the static PDF toward structured entity formats to maintain discoverability in AI-driven knowledge graphs [4][6]. The journal article is likely to persist as the primary unit of currency for academic promotion due to high institutional inertia in tenure committees. However, the delivery mechanism of the peer-review process may integrate structured transparency logs to counteract the opacity of AI-assisted drafting [5].

*Hypothesis: In the mid term, the focus of research publication management may move from the detection of synthetic text to the systematic verification of authorship and citation provenance through structured registries.*

### Far (2046–2051)

For research publications broadly, the long-term viability of the written manuscript as the primary research output remains unknown if the gap between automated generation [5] and detection [1] closes completely. One plausible outcome is a transition where the primary research output becomes a verifiable computational model or a registry-linked data stream [2], with written text serving only as a secondary, AI-generated summary for human consumption [4]. A rival outcome involves the erosion of trust in traditional publication management, leading to a fragmentation of the reward system as traditional journals fail to distinguish human discovery from synthetic synthesis. It remains unknown whether visibility infrastructure [4] can scale sufficiently to maintain a global, trusted record of discovery. The fundamental requirement for a verifiable and immutable record of intellectual priority resists change even at this horizon.

*Hypothesis A: In the far term, the primary research output may shift from the written manuscript to verifiable computational models or data streams linked to citation registries.*

*Hypothesis B (competing): In the far term, the inability to distinguish human from synthetic discovery* *may lead to a collapse of the traditional journal system and a fragmented, localized reward structure for research.*

## References


1. Dev To Ai (2026-04-06). “Will AI-Written Papers Be Detected? The Truth About AIGC Detection in 2026.” https://dev.to/yu_z_ed6c1c576593c5e7d41f/will-ai-written-papers-be-detected-the-truth-about-aigc-detection-in-2026-5b50
2. Dev To Ai (2026-04-06). “AI Citation Registries as Information Infrastructure for AI Systems.” https://dev.to/aigistry/ai-citation-registries-as-information-infrastructure-for-ai-systems-50o6
3. Lesswrong (2026-04-07). “An Alignment Journal: Features and policies.” https://www.lesswrong.com/posts/hvq7amw8FKZxEeKqA/an-alignment-journal-features-and-policies
4. Dev To Ai (2026-04-08). “Building AI Visibility Infrastructure: Inside Jonomor’s Architecture.” https://dev.to/jonomor_ecosystem/building-ai-visibility-infrastructure-inside-jonomors-architecture-3enp
5. Marktechpost (2026-04-09). “Google AI Research Introduces PaperOrchestra: A Multi-Agent Framework for Automated AI Research Paper Writing.” https://www.marktechpost.com/2026/04/08/google-ai-research-introduces-paperorchestra-a-multi-agent-framework-for-automated-ai-research-paper-writing/
6. Dev To Ai (2026-04-10). “How to Get Cited by AI Search Engines: ChatGPT, Perplexity & Claude in 2026.” https://dev.to/iris1031/how-to-get-cited-by-ai-search-engines-chatgpt-perplexity-claude-in-2026-21p0
7. Dev To Ai (2026-04-10). “Why DA Is Irrelevant for AI Citations (Data from 7 Site Audits).” https://dev.to/chudi_nnorukam/why-da-is-irrelevant-for-ai-citations-data-from-7-site-audits-1il0

## Research Training

*capability_id: research-training · created: 2026-05-21 · revised: 2026-05-21 · refs: 7*

### Near (2026–2031)

For research training broadly, the researcher training and development process may integrate AI-driven search and deep research tools to automate source identification and the generation of structured literature syntheses [3][4]. This modification changes the delivery mechanism of the literature review process rather than its educational substance. For the subsegment of researchers engaged in quantitative analysis, postgraduate research student skills development may incorporate AI-driven data analysis tools to automate dataset exploration and initial visualization [5]. Within the specific subsegment of machine learning research, student skills development may shift from a primary focus on theoretical mathematical derivation toward compute-intensive engineering and the management of large-scale empirical experiments [1]. This shift represents a change in educational substance, as research progress increasingly relies on compute scaling over mathematically principled architectures [1]. For technical subsegments requiring software development, skills development may pivot from teaching manual coding syntax to the orchestration of agentic AI tools and terminal-based AI agents [2][7]. In the subsegment of scientific writing and peer review training, the development of researcher skills may incorporate AI systems designed to generate constructive feedback based on author responses [6]. The institutional requirement for a written thesis and a formal oral defense is unlikely to change in the near term due to the high inertia of degree-awarding regulations.

*Hypothesis: In the near term, general research training will integrate AI-driven sourcing and synthesis, while technical subsegments will pivot toward compute engineering, agentic orchestration, and AI-mediated feedback loops.*

### Mid (2036–2041)

For the same broad and technical cohorts, the postgraduate research student skills development process may transition from a focus on information retrieval to the rigorous validation of AI-curated syntheses if current trajectories hold [3][4]. Extrapolating from the adoption of unified agentic systems, researcher training in technical subsegments may replace manual codebase maintenance with the orchestration of autonomous coding agents as a primary technical competency [2][7]. This shift may alter the substance of postgraduate research supervisor development, as supervisors may move from guiding manual methodology to validating agent-generated empirical workflows. The necessity for the researcher to frame a novel research question and maintain an original theoretical contribution is likely to persist unchanged because these tasks require high-level conceptual synthesis that current agentic trajectories do not replace.

*Hypothesis: In the mid term, research training will shift toward the validation of automated synthesis and the orchestration of agentic systems, shifting supervisory focus toward workflow validation.*

### Far (2046–2051)

For machine learning and technical research subsegments, the long-term trajectory of skills development depends on whether compute-intensive scaling continues to yield returns or reaches a threshold of diminishing returns [1]. If scaling remains the primary driver of discovery, research training may evolve into a specialized discipline of high-performance infrastructure management and agent orchestration. Conversely, if scaling returns diminish, a rival trajectory may see a renewed emphasis on mathematically principled architectures to enable new breakthroughs [1]. It remains unknown whether the human role in general researcher training will shift from primary synthesizer to an overseer of automated discovery systems [3][4]. The institutional requirement for independent peer-reviewed validation of research results will likely resist change even at this horizon to maintain social legitimacy.

*Hypothesis A: In the far term, research training in technical subsegments may fully transition into a specialized discipline of infrastructure management and agent orchestration.*

*Hypothesis B (competing): In the far term, research training in technical subsegments may return to a foundation of theoretical mathematics if compute-scaling returns diminish.*

## References


1. The Gradient (2024-11-16). "Shape, Symmetries, and Structure: The Changing Role of Mathematics in Machine Learning Research." https://thegradient.pub/shape-symmetry-structure/
2. Dev To Ai (2026-04-10). "Claude Code Cheatsheet: The Terminal AI Agent That Actually Codes." https://dev.to/mibii/claude-code-cheatsheet-the-terminal-ai-agent-that-actually-codes-36bk
3. OpenAI (2026-04-10). "ChatGPT for research." https://openai.com/academy/research
4. OpenAI (2026-04-10). "Research with ChatGPT." https://openai.com/academy/search-and-deep-research
5. OpenAI (2026-04-10). "Analyzing data with ChatGPT." https://openai.com/academy/data-analysis
6. Arxiv Cs Ai (2026-04-16). "GoodPoint: Learning Constructive Scientific Paper Feedback from Author Responses." https://arxiv.org/abs/2604.11924
7. Simon Willison (2026-04-25). "Quoting Romain Huet." https://simonwillison.net/2026/Apr/25/romain-huet/#atom-everything

## Strategy Management

*capability_id: strategy-management · created: 2026-06-03 · revised: 2026-06-03 · refs: 11*

### Near (2026–2031)

For institutions in China, state-funded research bodies, public universities subject to mandated viability reviews, and a subsegment of institutions adopting AI-driven synthesis and forecasting tools, strategy management is shifting toward a combination of external mandate alignment and tool-assisted synthesis. In China, national targets for AI deployment are being integrated directly into strategic plan development and strategic plan management via five-year planning cycles [1]. For state-funded research institutes, as demonstrated by the Alan Turing Institute, funding bodies may mandate revisions to strategic plan development to ensure adherence to legal duties and value-for-money metrics [2]. In public institutions facing enrollment and budget pressures, such as Iowa State University and East Carolina University, corporate performance management is increasingly driven by the use of viability data to execute program closures or mergers to meet specific budget reduction targets [6][10]. For institutions using automated synthesis tools, the processes of strategic reporting and business horizon scanning may be streamlined through platforms that generate competitive intelligence and market analysis [3][4][5], while specialized labor automation and AGI scenario analysis tools are being used to identify disruptive technological trends [7][11]. Simultaneously, an emerging shift in corporate performance management seeks to replace routine cognitive KPIs with metrics that reflect human-centric value [8], as executives increase AI budgets to move from urgency-driven to intent-driven investment [9]. These changes involve delivery-mechanism changes—specifically the automation of strategic reporting and horizon scanning—and changes to educational substance, particularly in how corporate performance management prioritizes fiscal viability [6][10] and redefined value metrics [8]. The fundamental legal fiduciary duties of governing boards and the statutory requirements of university charters are unlikely to change in this period.

*Hypothesis: In the near term, strategy management in these settings may be increasingly defined by the transition of AI from a technical project to an executive mandate, utilizing automated synthesis to accelerate reporting and the redefinition of performance metrics.*

### Mid (2036–2041)

For institutions in China, state-funded research bodies, public universities under state oversight, and adopters of AI strategy tools, if current trajectories hold, corporate performance management may shift toward the quantitative measurement of external benchmarks and AI-generated KPIs. Extrapolating from state and funder mandates [1][2] and mandated program reviews [6][10], strategic reporting could move from an internal reflection of institutional mission to a compliance verification tool, altering the educational substance of how institutional success is defined. For institutions using automated strategy platforms, the process of strategic plan development may shift from an act of primary synthesis to an act of auditing AI-generated outputs [3][4][5]. If the shift toward meaning-based KPIs persists [8], strategic vision development may diverge between institutions that prioritize AI-driven efficiency and those that prioritize human-centric value creation. The requirement for institutions to maintain a distinct institutional identity within their strategic vision development is likely to persist, though it will operate within boundaries set by external regulatory authorities and tool constraints.

*Hypothesis: In the mid term, corporate performance management in these segments may transition into a process of algorithmic compliance, where strategic plan development centers on the auditing of AI-synthesized trajectories.*

### Far (2046–2051)

For institutions in China, state-funded research bodies, public universities under state oversight, and adopters of AI strategy tools, the stability of strategic vision development remains unknown as the arrival of artificial general intelligence may decouple strategic planning from human temporal horizons [11]. One rival outcome suggests a total automation of corporate performance management where resource allocation is dynamically optimized by AI based on real-time labor market and viability signals. A competing outcome suggests a retreat to human-only deliberation for strategic vision development, driven by the need for social legitimacy and ethical accountability that AI cannot provide. The necessity for human governance boards to provide final approval for strategic plan management is likely to resist change due to the legal requirement for human accountability in institutional fiduciary duty.

*Hypothesis A: In the far term, strategy management may evolve into a continuous, automated optimization function where human roles are limited to the approval of AI-generated viability trajectories.*

*Hypothesis B (competing): In the far term, strategy management may revert to a purely human-centric deliberative process to preserve institutional legitimacy and navigate high-uncertainty "black swan" events that evade algorithmic forecasting.*

## References


1. Ai News (2026-04-02). "China's Five-Year Plan details the targets for AI deployment." https://www.artificialintelligence-news.com/news/chinas-five-year-plan-details-the-targets-for-ai-deployment/
2. Guardian Tech (2026-04-03). "UK's leading AI research institute told to make 'significant' changes." https://www.theguardian.com/technology/2026/apr/03/uk-alan-turing-institute-ai-research-underperforming-ukri
3. TechCrunch (2026-04-07). "AI startup Rocket offers vibe McKinsey-style reports at a fraction of the cost." https://techcrunch.com/2026/04/06/indian-startup-rocket-wants-its-ai-to-do-mckinsey-style-consulting-at-a-fraction-of-the-cost/
4. Dev To Ai (2026-04-07). "AI startup Rocket offers vibe McKinsey-style reports at a fraction of the cost." https://dev.to/minimal-architect/ai-startup-rocket-offers-vibe-mckinsey-style-reports-at-a-fraction-of-the-cost-1edd
5. Mit Sloan (2026-04-08). "Gain Consumer Insight With Generative AI." https://sloanreview.mit.edu/article/gain-consumer-insight-with-generative-ai/
6. Higher Ed Dive (2026-04-10). "Iowa State plans to cut or merge 23 programs after mandated review." https://www.highereddive.com/news/iowa-state-university-program-cuts-review-regents/817262/
7. Lesswrong (2026-04-21). "Introducing the Labor Automation Forecasting Hub." https://www.lesswrong.com/posts/So7wrd6djpQcoP4rz/introducing-the-labor-automation-forecasting-hub-1
8. Fast Company (2026-04-23). "The Age of AI means we need to throw out our old KPIs and replace them with new ones." https://www.fastcompany.com/91521422/ai-kpis-new-metrics?partner=rss&utm_source=rss&utm_medium=feed&utm_campaign=rss+fastcompany&utm_content=rss
9. Dzone Ai (2026-04-24). "88% of Executives Plan to Increase AI Budgets by 2026 - Are We Investing With Intent or Just Urgency?" https://feeds.dzone.com/link/23558/17325011/increasing-ai-budgets-investing-intent-or-urgency
10. Higher Ed Dive (2026-04-27). "East Carolina University plans to cut 44 academic programs." https://www.highereddive.com/news/east-carolina-university-cut-around-44-academic-programs/818622/
11. Arxiv Cs Cy (2026-04-28). "Artificial General Intelligence Forecasting and Scenario Analysis: State of the Field, Methodological Gaps, and Strategic Implications." https://arxiv.org/abs/2604.22766

## Student Administration

*capability_id: student-administration · created: 2026-04-11 · revised: 2026-04-11 · refs: 1*

### Near (2026–2031)

For public universities subject to state-mandated program reviews, as demonstrated by the experience at Iowa State University [1], the process of programme transfer management may face sudden surges in administrative volume. This occurs because mandated closures of low-enrollment programs require the systematic reassignment of currently enrolled students to merged or alternative programs [1]. This represents a change to the delivery mechanism of student administration—specifically the volume and speed of processing—rather than a change in the educational substance of the capability. Student records & details management will likely remain stable in its core function, though the frequency of updates to degree paths will increase. It is unlikely that the fundamental regulatory requirements for degree conferral will change yet, as these remain bound by external accreditation standards.

*Hypothesis: In the near term, for a subset of public institutions facing state-mandated program cuts, programme transfer management may become a significant administrative bottleneck.*

### Mid (2036–2041)

For public universities facing similar regulatory pressures, if current trajectories of mandated program reviews hold, the increased load on programme transfer management could lead to the adoption of automated matching tools to suggest alternative programs for displaced students. This would further shift the delivery mechanism of student administration toward algorithmic decision-support. Extrapolating from current administrative constraints, student financial administration may require tighter integration with transfer workflows to manage the reallocation of program-specific scholarships. However, the final validation of enrolment status management will likely persist as a human-governed process to ensure legal compliance with state funding mandates.

*Hypothesis: In the mid term, for institutions experiencing frequent program restructuring, the delivery mechanism of programme transfer management may shift toward automated suggestion systems.*

### Far (2046–2051)

For institutions subject to state-level programmatic volatility, one plausible outcome is that programme transfer management becomes a dynamic process that re-aligns student enrolments in real-time based on labor market signals and state mandates. A rival outcome is that the persistent instability of program offerings leads to a shift toward broader, degree-agnostic enrolment status management, which would reduce the frequency of specific programme transfer events. It remains unknown whether state regents will maintain the authority to mandate such cuts or if institutional autonomy over program design will be restored. The legal requirement for permanent, immutable student records will likely resist change even at this horizon.

*Hypothesis A: In the far term, for public universities in volatile regulatory environments, programme transfer management may be integrated into dynamic labor-market alignment systems.*

*Hypothesis B (competing): In the far term, the administrative burden of frequent program cuts may drive a shift toward modular, degree-agnostic enrolment status management, reducing the need for traditional programme transfer processes.*

### References


1. Higher Ed Dive (2026). "Iowa State plans to cut or merge 23 programs after mandated review." https://www.highereddive.com/news/iowa-state-university-program-cuts-review-regents/817262/

## Student Admission Management

*capability_id: student-admission-management · created: 2026-04-25 · revised: 2026-04-25 · refs: 1*

### Near (2026–2031)

For a subsegment of institutions where financial aid offer letters are characterized by opacity and confusing terminology [1], the process of offer, acceptance & quota management may be impacted by a shift in how students decode these documents. Because the current delivery mechanism of financial aid communication often omits critical cost information [1], students may increasingly employ large language models to normalize and compare disparate offer formats. This represents a change in the delivery mechanism of the capability, as the point of interpretation shifts from the institution's intent to an AI-mediated analysis. However, the student application management process is unlikely to change in the near term due to the institutional inertia of established admissions cycles and regulatory reporting requirements.

*Hypothesis: In the near term, the offer, acceptance & quota management process for a subsegment of institutions may see a shift in power as students use AI to decode opaque financial aid offers.*

### Mid (2036–2041)

Extrapolating from the current lack of transparency in offer letters [1], some institutions may attempt to use AI-driven personalization to optimize quota management by tailoring the presentation of financial aid to specific student psychographics. This shift in delivery mechanism could increase the risk of counterproductivity, where the pursuit of higher yield rates through calculated opacity undermines the social legitimacy of the admission process. Despite these potential shifts, the legal frameworks governing federal financial aid reporting will likely remain a stabilizing constraint, preventing the total removal of standard disclosures.

*Hypothesis: In the mid term, some institutions may use AI to further personalize financial aid offers to optimize yield, potentially increasing the opacity of the offer, acceptance & quota management process.*

### Far (2046–2051)

In the far term, two rival trajectories emerge for this subsegment of institutions. One path involves a regulatory response to AI-driven opacity, mandating a standardized, machine-readable format for all financial aid offers to ensure transparency. A competing path involves the obsolescence of the "offer letter" as a discrete artifact, replaced by real-time, AI-managed cost-of-attendance dashboards. It remains unknown whether accreditation bodies will view AI-generated personalized pricing as a violation of equitable access standards. The fundamental requirement for a legal offer of admission will likely resist change due to the contractual nature of enrollment.

*Hypothesis A: In the far term, regulatory mandates may force a transition to standardized, AI-verifiable financial aid disclosures.*

*Hypothesis B (competing): In the far term, the traditional offer letter may be replaced by dynamic, AI-managed financial transparency dashboards.*

### References


1. The Hechinger Report (2026). "How much will that college cost you? Good luck figuring it out." https://hechingerreport.org/how-much-will-that-college-cost-you-good-luck-figuring-it-out-2/

## Student Assessment

*capability_id: student-assessment · created: 2026-06-01 · revised: 2026-06-01 · refs: 18*

### Near (2026–2031)

For student assessment broadly, the educational substance is shifting from the production of static academic artifacts toward the evaluation of learning processes [23]. In the process of assessment delivery, students are increasingly utilizing iterative human-AI writing cycles [1], a transition that highlights systemic flaws in the reliance on essay assessments within mass education systems [3] and introduces the risk of "cognitive surrender," where students may abandon logical thinking in favor of synthetic output [5]. This shift in delivery mechanism may lead to a homogenization of student writing and thinking patterns, which complicates assessment marking and feedback by eroding the distinct critical voice [9]. In the process of assessment administration, the reliability of authorship verification is constrained by the persistence of false positives in AI detectors [13] and the continuous decay of detection pipelines as models evolve [28], while "humanizer" tools further erode the utility of sentence-level detection [15]. For institutions implementing automated feedback, the process of assessment marking and feedback is subject to algorithmic bias, with evidence that AI may provide disparate levels of praise and criticism based on student race [27]. For a subset of institutions, marking and feedback now includes the automation of mock exams [8] and the use of confidence-based cascade scoring for routine grading [21]. High-stakes, invigilated examinations are unlikely to change in the near term due to professional accreditation requirements and the institutional need for authenticated baseline competency [19].

*Hypothesis: In the near term, assessment delivery may shift from artifact-production to process-evaluation, while assessment administration is increasingly constrained by the technical failure of authorship detection tools.*

### Mid (2036–2041)

For student assessment broadly, if current trajectories hold, the educational substance of assessment delivery may shift from evaluating a final product to evaluating a student's capacity to audit, verify, and correct synthetic errors [5][7]. This represents a shift in substance from measuring output to measuring the ability to steer and validate synthetic agents [14]. To maintain the validity of assessment results management, institutions may implement deliverable-oriented governance frameworks to manage the inherent opacity of these AI systems [20]. For a subset of institutions focusing on multimodal pedagogical analysis, results management could incorporate the longitudinal mapping of cognitive effort via educational fMRIs [4]. The basic administrative recording of credits and formal result transmission will likely persist unchanged due to high regulatory inertia.

*Hypothesis: In the mid term, the substance of student assessment may shift from measuring product output to measuring a student's ability to audit and validate synthetic agents.*

### Far (2046–2051)

For student assessment broadly, the long-term trajectory depends on whether synthetic protocols can reliably replace human judgment in certifying higher-order cognitive capabilities [18]. It remains unknown if LLM-based protocols can meaningfully measure collaboration, creativity, and critical thinking without reducing these capabilities to observable patterns [17]. Certain core requirements of professional certification will likely resist full automation to maintain social legitimacy and institutional trust. A primary uncertainty is whether the trend of "cognitive surrender" persists or whether institutions can enforce a return to intellectual agency [5]. Rival outcomes emerge based on the tension between the efficiency of synthetic certification and the social value of human-verified expertise.

*Hypothesis A: In the far term, synthetic protocols may become the primary mechanism for certifying higher-order cognitive capabilities.*

*Hypothesis B (competing): In the far term, high-stakes assessments may return to strictly analog, human-*

*verified environments to maintain social legitimacy and counter cognitive surrender.*

## References


1. The Conversation (2026-03-26). "College students are writing with AI – but a pilot study finds they're not simply letting it write for them." https://theconversation.com/college-students-are-writing-with-ai-but-a-pilot-study-finds-theyre-not-simply-letting-it-write-for-them-276856
2. The Conversation Au (2026-04-01). "Does AI mean more uni students are plagiarising their work?" https://theconversation.com/does-ai-mean-more-uni-students-are-plagiarising-their-work-279565
3. Wonkhe (2026-04-01). "Generative AI exposes longstanding flaws in the use of essay assessments in a mass system." https://wonkhe.com/blogs/generative-ai-exposes-longstanding-flaws-in-the-use-of-essay-assessments-in-a-mass-system/
4. Getting Smart (2026-04-01). "Educational fMRIs: Dynamic Pedagogy and Pedagogical Analysis in the Multimodal AI Era." https://www.gettingsmart.com/2026/04/01/educational-fmris-dynamic-pedagogy-and-pedagogical-analysis-in-the-multimodal-ai-era/
5. Ars Technica (2026-04-03). ""Cognitive surrender" leads AI users to abandon logical thinking, research finds." https://arstechnica.com/ai/2026/04/research-finds-ai-users-scarily-willing-to-surrender-their-cognition-to-llms/
6. Dev To Ai (2026-04-06). "Will AI-Written Papers Be Detected? The Truth About AIGC Detection in 2026." https://dev.to/yu_z_ed6c1c576593c5e7d41f/will-ai-written-papers-be-detected-the-truth-about-aigc-detection-in-2026-5b50
7. Dev To Ai (2026-04-07). "Where Does Truth Live in AI-Generated Code?" https://dev.to/wintrover/where-does-truth-live-in-ai-generated-code-5hmc
8. BBC News (2026-04-07). "'Our teachers use AI to mark mock exams'." https://www.bbc.com/news/articles/c6207xenj1do?at_medium=RSS&at_campaign=rss
9. Hacker News (2026-04-07). "AI may be making us think and write more alike." https://dornsife.usc.edu/news/stories/ai-may-be-making-us-think-and-write-more-alike/
10. Plos One (2026-04-07). "Correction: Evaluating cognitive depth of AI-generated multiple-choice questions with Bloom's Taxonomy." https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0346800
11. Arxiv Cs Cy (2026-04-08). "A Multi-Agent Approach to Validate and Refine LLM-Generated Personalized Math Problems." https://arxiv.org/abs/2604.05160
12. Dev To Ai (2026-04-08). "Your article is well-written. But is it yours?" https://dev.to/dzhuneyt/your-article-is-well-written-but-is-it-yours-59kh
13. Dev To Ai (2026-04-09). "Why AI Detectors Produce False Positives: A Technical Analysis." https://dev.to/laakash/why-ai-detectors-produce-false-positives-a-technical-analysis-2gpc
14. Atlantic (2026-04-10). "This Is What Fully Automated School Looks Like." https://www.theatlantic.com/technology/2026/04/ai-agents-school-education/686754/?utm_source=feed
15. Dev To Ai (2026-04-11). "I Built a Free AI Detector + Humanizer with Sentence-Level Highlighting." https://dev.to/fafa_ai/i-built-a-free-ai-detector-humanizer-with-sentence-level-highlighting-51jj
16. Getting Smart (2026-04-13). "A Call to Action for AI to Promote Mathematical Reasoning." https://www.gettingsmart.com/2026/04/13/a-call-to-action-for-ai-to-promote-mathematical-reasoning/
17. Marktechpost (2026-04-14). "Google AI Research Proposes Vantage: An LLM-Based Protocol for Measuring Collaboration, Creativity, and Critical Thinking." https://www.marktechpost.com/2026/04/13/google-ai-research-proposes-vantage-an-llm-based-protocol-for-measuring-collaboration-creativity-and-critical-thinking/
18. Arxiv Cs Cy (2026-04-14). "Explainability and Certification of AI-Generated Educational Assessments." https://arxiv.org/abs/2604.09622
19. University Affairs (2026-04-14). "Reconsidering the value of exams in the age of AI." https://universityaffairs.ca/career-advice/reconsidering-the-value-of-exams-in-the-age-of-ai/
20. Arxiv Cs Ai (2026-04-23). "AI to Learn 2.0: A Deliverable-Oriented Governance Framework and Maturity Rubric for Opaque AI in Learning-Intensive Domains." https://arxiv.org/abs/2604.19751

21. Arxiv Cs Cy (2026-04-23). “Do Small Language Models Know When They’re Wrong? Confidence-Based Cascade Scoring for Educational Assessment.” https://arxiv.org/abs/2604.19781
22. Elearning Industry (2026-04-22). “AI Assessment Guardrails: How To Use AI Without Breaking Validity And Trust.” https://elearningindustry.com/ai-assessment-guardrails-how-to-use-ai-without-breaking-validity-and-trust
23. EDUCAUSE Review (2026-04-23). “Teaching and Learning with AI: Three Priorities.” https://er.educause.edu/podcasts/educause-exchange/teaching-and-learning-with-ai-three-priorities
24. The Hechinger Report (2026-04-27). “AI gives more praise, less criticism to Black students.” https://hechingerreport.org/proof-points-ai-bias-feedback/
25. Dev To Ai (2026-04-27). “LLM Drift: Why Your AI Detection Pipeline is Quietly Decaying (Kimi K2 Benchmark).” https://dev.to/thoams_aidetection/llm-drift-why-your-ai-detection-pipeline-is-quietly-decaying-kimi-k2-benchmark-3gml

## Student Attraction & Recruitment

*capability_id: student-attraction-recruitment · created: 2026-05-27 · revised: 2026-05-27 · refs: 8*

### Near (2026–2031)

For a subset of institutions adopting AI-driven marketing tools and those in regions experiencing systemic enrollment shifts, the process of prospective student engagement may shift as discovery moves from keyword-based search to AI-mediated platforms, requiring a transition from traditional SEO to AI search optimization to maintain visibility [1][6]. For institutions attending recruitment fairs, the process of prospective student engagement may incorporate AI to convert raw lead data into qualified narratives [4], automate post-event follow-up [7], and implement AI-driven lead scoring to prioritize high-value prospects [8]. Simultaneously, the production cost of personalized visual content for domestic and international student recruitment may decrease through the adoption of AI-driven video generation tools [2]. In regions where COVID-era remote learning has created a long-term drain on college enrollment, the process of domestic student recruitment is constrained by a shrinking pool of traditional candidates [3]. Within the process of scholarship and bursary management, a lack of cost transparency in financial aid offers creates significant friction during the final stages of prospective student engagement [5]. These changes primarily affect the delivery mechanism of recruitment—specifically lead qualification and content production—rather than the educational substance of the degree programs. The internal fiduciary and audit logic used to allocate scholarship funds is unlikely to change in this period due to rigid regulatory and institutional constraints.

*Hypothesis: In the near term, a subset of institutions may adjust prospective student engagement to prioritize AI search optimization and automated lead scoring pipelines, while regional enrollment declines and communication failures in scholarship management may constrain overall recruitment effectiveness.*

### Mid (2036–2041)

For the same subset of institutions utilizing AI-driven engagement, if current trajectories hold, the process of prospective student engagement could shift toward the maintenance of structured data schemas designed for machine readability rather than human-centric web narratives [1][6]. Extrapolating from current lead-intelligence deployments, the qualification and scoring of recruitment fair data may move from an asynchronous post-event task to a real-time automated pipeline [4][7][8]. The proliferation of low-cost, AI-generated video may lead to a saturation of personalized outreach, potentially shifting the focus of international student recruitment toward the verification of institutional authenticity and human-led validation [2]. If regional enrollment drains persist, domestic student recruitment may be forced to pivot toward alternative lead-generation mechanisms that bypass traditional pipeline markers [3]. To resolve existing friction in scholarship and bursary management, institutions may deploy AI-mediated transparency tools to translate complex financial aid terms into personalized cost projections for families [5]. However, the institutional requirement for faculty-led program descriptions will likely persist as a core requirement for academic legitimacy.

*Hypothesis: In the mid term, prospective student engagement for some institutions may shift toward the production of machine-optimized entity data, while AI may be used to reduce information asymmetry in scholarship and bursary management.*

### Far (2046–2051)

For institutions relying on AI-mediated prospective student engagement, it remains unknown whether AI search platforms will maintain neutral recommendation architectures or implement proprietary visibility constraints. The mechanism of domestic student recruitment may either bifurcate into hyper-optimized algorithmic targeting or revert to high-touch, human-centric networking as a reaction to AI content saturation [1][6]. One unresolvable uncertainty is whether the automation of lead scoring and qualification will eventually erode the social legitimacy of the recruitment process by removing human judgement from the initial selection of candidates [8]. Despite these shifts, the basic institutional requirement for legal accreditation and the verification of degree-awarding status will resist change even at this horizon.

*Hypothesis A: In the far term, recruitment for some institutions becomes a function of algorithmic visibility management, where institutional "attractiveness" is defined by alignment with AI recommendation parameters.*

*Hypothesis B (competing): In the far term, the saturation of AI-generated recruitment content triggers a systemic return to localized, human-mediated networks to establish trust and authenticity in student recruitment.*

## References


1. Dev To Ai (2026-04-06). "What is the difference between AI content and AI search optimised content?" https://dev.to/rank4ai/what-is-the-difference-between-ai-content-and-ai-search-optimised-content-37bc
2. Google AI Blog (2026-04-02). "Create, edit and share videos at no cost in Google Vids" https://blog.google/products-and-platforms/products/workspace/google-vids-updates-lyria-veo/
3. Higher Ed Dive (2026-04-10). "COVID remote learning put drain on college enrollment" https://www.highereddive.com/news/covid-remote-learning-put-drain-on-college-enrollment/816899/
4. Dev To Ai (2026-04-12). "From Notes to Narrative: How AI Automates Trade Show Lead Intelligence" https://dev.to/ken_deng_ai/from-notes-to-narrative-how-ai-automates-trade-show-lead-intelligence-1239
5. The Hechinger Report (2026-04-20). "How much will that college cost you? Good luck figuring it out" https://hechingerreport.org/how-much-will-that-college-cost-you-good-luck-figuring-it-out-2/
6. Dev To Ai (2026-04-25). "I Audited a Business's AI Visibility Across Four Platforms. The Results Were Worse Than Expected." https://dev.to/edvisageglobal/i-audited-a-businesss-ai-visibility-across-four-platforms-the-results-were-worse-than-expected-4ml5
7. Dev To Ai (2026-04-26). "Stop Chasing Ghosts: Automate Your Post-Trade Show Follow-Up" https://dev.to/ken_deng_ai/stop-chasing-ghosts-automate-your-post-trade-show-follow-up-5f65
8. Dev To Ai (2026-04-27). "Instant Lead Scoring: Teaching AI to Sort Trade Show Prospects" https://dev.to/ken_deng_ai/instant-lead-scoring-teaching-ai-to-sort-trade-show-prospects-4l1l

# Student Enrolment

*capability_id: student-enrolment · created: 2026-04-10 · revised: 2026-04-10 · refs: 1*

## Near (2026–2031)

In the United States, programme enrolment volumes for first-year students have decreased [1]. This decline is linked to a lower rate of completion for financial aid applications (FAFSA) and standardized testing (ACT) [1]. These frictions at the entry point of the programme enrolment process reduce the pool of eligible candidates entering the pipeline [1]. Because this drain occurs at the initial sign-up phase, module selection management is unlikely to change in the near term as it is a downstream process. Student induction management remains largely unaffected by these volume shifts in the immediate term [1].

*Hypothesis: In the near term, programme enrolment in the US region may experience continued volatility as institutions respond to changes in student application behaviors.*

## Mid (2036–2041)

In the United States, if current trajectories of declining first-year sign-ups hold, institutions may alter the delivery mechanism of programme enrolment to reduce friction [1]. This could involve the deployment of automated tools to assist students in navigating FAFSA and ACT requirements, shifting the process from a passive administrative task to an active, supported workflow. Extrapolating from the volume drain, institutions may also modify student induction management to prioritize retention of a smaller entering cohort [1]. The fundamental regulatory requirements for financial aid eligibility will likely persist unchanged due to government oversight.

*Hypothesis: In the mid term, institutions in the affected region may adopt more automated delivery mechanisms for programme enrolment to mitigate volume loss.*

## Far (2046–2051)

In the United States, the long-term elasticity of student demand relative to remote learning formats remains unknown [1]. If the drain on traditional programme enrolment persists, the capability may diverge into two rival outcomes. One outcome involves the complete automation of the enrolment pipeline, where institutional barriers are minimized to maximize throughput. A competing outcome involves a return to high-touch, human-centric enrolment processes to restore social legitimacy and perceived value [1]. Legal requirements for institutional accreditation will resist change even at this horizon, as they provide the basis for enrolment legitimacy.

*Hypothesis A: In the far term, programme enrolment may transition to a continuous, fully automated delivery mechanism to minimize student friction.*

*Hypothesis B (competing): In the far term, institutions may shift toward high-touch, human-led enrolment processes to counteract the perceived depersonalization of remote learning.*

## References


1. Higher Ed Dive (2026). "COVID remote learning put drain on college enrollment." https://www.highereddive.com/news/covid-remote-learning-put-drain-on-college-enrollment/816899/

## Student Support & Wellbeing Management

*capability_id: student-support-wellbeing-management · created: 2026-05-15 · revised: 2026-05-15 · refs: 10*

### Near (2026–2031)

For a subset of institutions and specific student populations incorporating AI-mediated workflows, shifts are emerging in both the delivery and substance of support processes. In the process of personal learning management, the delivery mechanism is shifting toward AI-automated synthesis of study materials through the deployment of tools such as Adobe Acrobat Spaces [4]. Within academic skill development, the educational substance is shifting toward the regulation of "cognitive partnership," as students move from simple generation toward iterative brainstorming [1] and the adoption of frameworks like TACO to prevent AI from substituting for logical thinking [8]. This shift is a response to the risk of "cognitive surrender," where users abandon independent reasoning [2] and exhibit linguistic homogenization in their thought and writing patterns [5]. In the process of disability support management, delivery is shifting toward the use of AI for drafting Individualized Education Programs (IEPs) and coursework support [9], though these deployments currently lack systematic safeguarding and policy frameworks [9]. Within student health & wellbeing, delivery is incorporating AI despite workforce tension between clinical enthusiasm and fear of displacement [3], while the substance of triage is adapting to address critical safety failures in LLMs handling suicide and self-harm crises [10] and risks of AI-induced delusions [6]. In the region of Florida, this process is being formally constrained by state-directed partnerships to shield families from AI harm [7]. The processes of student financial advice and housing advice are unlikely to change in this horizon because they remain tied to external regulatory frameworks and static administrative constraints.

*Hypothesis: In the near term, academic skill development for a subset of students may transition toward the formal regulation of cognitive partnership to mitigate cognitive surrender, while student health & wellbeing may prioritize the mitigation of critical AI failure modes in crisis triage.*

### Mid (2036–2041)

For the same subset of institutions and regions, if current trajectories hold, the academic skill development process may formalize the auditing of machine-generated logical structures as a core competency, extrapolating from early frameworks that distinguish between cognitive support and cognitive substitution [8]. Within disability support management, the implementation of "Responsible Inference Engines" may evolve from a policy recommendation into a standard operational requirement for safeguarding students with learning differences [9]. In student health & wellbeing, AI-driven documentation and triage may further integrate into the workforce, provided that institutional protocols resolve the tension between professional fear and operational efficiency [3] and establish redundant human-in-the-loop overrides for high-risk mental health interventions [10]. In regions implementing standardized harm reporting, such as Florida, these mechanisms may evolve into permanent regulatory requirements for student wellbeing management [7]. The human-led provision of personal tutors is likely to persist unchanged as the primary mechanism for emotional scaffolding and motivational support, as these require a level of social legitimacy and human empathy not evidenced in current AI deployments.

*Hypothesis: In the mid term, institutions supporting AI-integrated workflows may establish formalized standards for the regulation of cognitive partnership and the safe deployment of inference engines for disability support.*

### Far (2046–2051)

For the subsegments of support focused on literacy, disability, and mental health, long-term outcomes diverge based on the resolution of cognitive and safety risks. One trajectory suggests a stabilized hybrid ecosystem where "cognitive partnership" is a foundational literacy and disability support is managed by validated, responsible inference engines [8][9]. A competing trajectory suggests a structural retrenchment toward human-only "safe zones" for wellbeing and academic skill development, driven by persistent systemic “cognitive surrender” [2] and the inability of AI to reliably manage high-stakes mental health crises [10]. A primary unknown remains whether the linguistic homogenization observed in early AI adoption [5] leads to a permanent erosion of divergent thinking across the student population. Despite these shifts, the final legal and ethical accountability for high-stakes student health outcomes will likely resist automation due to the necessity of professional licensure and institutional liability.

*Hypothesis A: In the far term, student support may evolve into a highly stratified system where AI manages low-stakes synthesis and administrative disability support, while high-stakes wellbeing is strictly human-gated.*

*Hypothesis B (competing): In the far term, pervasive cognitive surrender may force a systemic return to analogue academic skill development and wellbeing processes to preserve human agency and safety.*

## References


1. The Conversation (2026-03-26). “College students are writing with AI – but a pilot study finds they’re not simply letting it write for them.” https://theconversation.com/college-students-are-writing-with-ai-but-a-pilot-study-finds-theyre-not-simply-letting-it-write-for-them-276856
2. Ars Technica (2026-04-03). ““Cognitive surrender” leads AI users to abandon logical thinking, research finds.” https://arstechnica.com/ai/2026/04/research-finds-ai-users-scarily-willing-to-surrender-their-cognition-to-llms/
3. NPR (2026-04-07). “AI in the mental health care workforce is met with fear, pushback — and enthusiasm.” https://www.npr.org/2026/04/07/nx-s1-5771707/mental-health-care-workforce-artificial-intelligence-ai
4. TechCrunch (2026-04-07). “Adobe launches Acrobat Spaces, a free AI-powered study tool for students.” https://techcrunch.com/2026/04/07/adobe-launches-acrobat-spaces-a-free-ai-powered-study-tool-for-students/
5. Hacker News (2026-04-07). “AI may be making us think and write more alike.” https://dornsife.usc.edu/news/stories/ai-may-be-making-us-think-and-write-more-alike/
6. Daily Ai (2025-05-09). “ChatGPT Is Making People Think They’re Gods and Their Families Are Terrified.” https://dailyai.com/2025/05/chatgpt-is-making-people-think-theyre-gods-and-their-families-are-terrified/
7. Future Of Life (2026-03-09). “Governor DeSantis Directs Florida State Agencies to Partner with Future of Life Institute to Shield Families from AI Harm.” https://futureoflife.org/press-release/desantis-directs-florida-agencies-to-partner-with-fli/
8. Arxiv Cs Cy (2026-04-22). “Students Know AI Should Not Replace Thinking, but How Do They Regulate It? The TACO Framework for Human-AI Cognitive Partnership.” https://arxiv.org/abs/2604.18737
9. Getting Smart (2026-04-22). “Responsible Inference Engines: Safeguarding Students with Learning Differences in the AI Era.” https://www.gettingsmart.com/2026/04/22/responsible-inference-engines-safeguarding-students-with-learning-differences-in-the-ai-era/
10. Fast Company (2026-04-24). “LLMs don’t get mental health right. We need a two-pronged approach to fix them.” https://www.fastcompany.com/91527462/llms-dont-get-mental-health-right-we-need-a-two-pronged-approach-to-fix-them-ai-suicide-self-harm?partner=rss&utm_source=rss&utm_medium=feed&utm_campaign=rss+fastcompany&utm_content=rss

## Supporting Services

*capability_id: supporting-services · created: 2026-05-11 · revised: 2026-05-11 · refs: 3*

### Near (2026–2031)

For specific subsegments of supporting services, primarily food and retail management and venue management, the delivery mechanisms for logistics and monitoring are shifting toward automated digital systems. In food and retail management, the implementation of automated approval systems for supply chain logistics creates a dependency where physical goods cannot move without digital validation, which may lead to the physical waste of perishables when systems fail to approve shipments [1]. In venue management, the integration of AI video analytics transforms traditional CCTV into intelligent monitoring systems capable of detecting real-time risks through object detection and tracking [3]. Furthermore, the application of predictive intelligence allows these venues to shift from reactive observation toward forecasting movement patterns and density changes [2]. Processes such as religious support and child care management are unlikely to adopt these specific automated frameworks in the near term because their institutional value is derived from human presence and social legitimacy.

*Hypothesis: In the near term, food and retail management may experience increased operational fragility due to digital validation failures, while venue management may shift toward proactive risk detection and predictive spatial monitoring.*

### Mid (2036–2041)

For the food, retail, and venue management subsegments, if current trajectories of digital integration hold, the stability of these services will depend on the resilience of the underlying software. Extrapolating from current failures in automated approvals, the risk of systemic waste in food and retail management could become a structural vulnerability unless institutions implement manual override protocols [1]. In venue management, predictive intelligence and video analytics may evolve from monitoring density to automating real-time resource allocation and crowd control mechanisms [2][3]. However, the educational substance of complaint and compliment management is likely to remain unchanged, as the resolution of human grievances typically requires social empathy and discretionary authority.

*Hypothesis: In the mid term, food logistics may face persistent risks of "digital waste," while venue management may integrate predictive analytics into automated operational resource allocation.*

### Far (2046–2051)

For the food, retail, and venue management subsegments, the long-term outcome depends on whether the industry optimizes for centralized efficiency or distributed resilience. One possibility involves the development of high-resilience, autonomous validation systems that mitigate the single points of failure currently causing supply chain waste [1]. A rival outcome is a state of chronic operational instability where the gap between digital approval and physical reality creates permanent inefficiencies in food and retail management. It remains unknown if the cumulative cost of system-induced waste will force a return to non-digital, localized procurement. Processes like religious support will likely resist this trajectory throughout this horizon as their primary function is the provision of human-centric spiritual care.

*Hypothesis A: In the far term, these subsegments may achieve autonomous, high-resilience operational states via distributed intelligence and predictive validation.*

*Hypothesis B (competing): In the far term, supporting services may experience structural degradation and chronic waste due to an irreversible dependence on rigid, centralized digital approval mechanisms.*

### References


1. Science Daily Comp (2026-04-05). "Truckloads of food are being wasted because computers won't approve them." https://www.sciencedaily.com/releases/2026/04/260403224505.htm
2. Dev To Ai (2026-04-22). "Architecting Predictive Intelligence for Smart Venues." https://dev.to/t

anisha_khiara_4b390b7946/architecting-predictive-intelligence-for-smart-venues-1lkj
3. Dev To Ai (2026-04-23). “Integrating AI Video Analytics into Venue Operations.” https://dev.to/tanisha_khiara_4b390b7946/integrating-ai-video-analytics-into-venue-operations-2m09

## Teaching & Learning Delivery

*capability_id: teaching-learning-delivery · created: 2026-06-02 · revised: 2026-06-02 · refs: 20*

### Near (2026–2031)

For teaching & learning delivery broadly, the educational substance is shifting toward skills-based learning frameworks to address workplace volatility [13] and a prioritization of learning processes over the production of academic artifacts [19]. Within teaching & learning content management, the delivery mechanism is shifting from manual authoring to automated multimodal generation, including instructional videos [6], presentations derived from URLs [16], automated multimodal illustrations [7], and localized materials for Career and Technical Education (CTE) [1]. For institutions adopting disciplinary-specific AI integration, as demonstrated at the University of Surrey, content management is shifting from general AI literacy toward embedding AI use within discipline-specific standards [24]. This process is further modified by the transition to agentic co-authoring tools that modify documents during the drafting phase [20] and the use of expressive synthetic speech to reduce reliance on human voice-overs in asynchronous materials [21]. In non presence based teaching, delivery mechanisms are being modified for specific subsegments by AI avatars that convert static documents into interactive experiences [14], personalized practice systems that maintain learning rate regularity at scale [3], and specialized AI tutoring for coding [2], music exam preparation [4], math problem generation [8], and low-resource languages [22]. For institutions implementing structured AI integration in software engineering and STEM, presence based teaching is being modified by frameworks that establish a "cognitive partnership" to ensure AI does not replace the student's internal thinking process [15][17]. The core social mediation and professional socialization functions of presence based teaching are unlikely to change yet because the social legitimacy of the human instructor remains a primary requirement for institutional accreditation.

*Hypothesis: In the near term, the capability shifts toward a process-oriented educational substance supported by a delivery mechanism of automated multimodal content and adaptive non presence based interfaces.*

### Mid (2036–2041)

For institutions adopting automated delivery models and STEM-focused subsegments, the delivery mechanism of non presence based teaching may shift toward agent-led orchestration of student pathways if current trajectories in AI tutoring and autonomous schooling hold [2][10]. Within teaching & learning content management, the process may evolve from a creation-centric activity into a verification-heavy auditing role, provided that governance frameworks for quality, accuracy, and trust are standardized [5][11]. If the risk of "cognitive bypass"—where students use AI as a substitute for critical thinking—persists, presence based teaching could shift toward constrained delivery models that intentionally limit AI autonomy to protect the cognitive substance of the learning process [17]. The requirement for human-led mentorship in high-stakes professional socialization is likely to persist because it is embedded in the regulatory regimes of professional licensing and accreditation.

*Hypothesis: In the mid term, content management may transition into a verification and governance function, while non presence based teaching may shift toward agent-orchestrated pathways.*

### Far (2046–2051)

For the same subset of institutions, the delivery of teaching and learning may diverge based on how the tension between efficiency and cognitive development is resolved. One outcome involves the full transition of non presence based teaching to autonomous agent-led systems for knowledge acquisition, leaving human instructors to focus exclusively on high-level synthesis and professional identity [10]. A competing outcome involves a strategic re-centering on high-touch, human-only presence based teaching as a prestige marker for elite accreditation, effectively bifurcating the capability. It remains unknown whether disciplinary standards [24] will evolve to incorporate AI as a foundational cognitive tool or if they will move toward "AI-free" zones to preserve human reasoning. The core requirement for a certified human authority to validate a learner's professional competence is likely to resist change even at this horizon due to the social nature of professional legitimacy.

*Hypothesis A: In the far term, the capability bifurcates into autonomous agent-led knowledge acquisition and high-touch human-led synthesis.*

*Hypothesis B (competing): In the far term, the capability reverts to human-centric presence based teaching as a primary mechanism for maintaining the value of professional accreditation.*

## References


1. Elearning Industry (2026-04-05). "Leveraging AI And Automation To Streamline CTE Content Creation And Localization." https://elearningindustry.com/leveraging-ai-and-automation-to-streamline-cte-content-creation-and-localization
2. The Hechinger Report (2026-04-06). "The quest to build a better AI tutor." https://hechingerreport.org/proof-points-ai-tutor-python/
3. Arxiv Cs Cy (2026-04-07). "Personalized AI Practice Replicates Learning Rate Regularity at Scale." https://arxiv.org/abs/2604.03246
4. Dev To Ai (2026-04-07). "Automating Excellence: AI for Custom Music Exam & Recital Prep." https://dev.to/ken_deng_ai/automating-excellence-ai-for-custom-music-exam-recital-prep-2e2d
5. Elearning Industry (2026-04-07). "A 4-Step Framework For Using AI Transparently In Educational Content." https://elearningindustry.com/a-4-step-framework-for-using-ai-transparently-in-educational-content
6. Google AI Blog (2026-04-02). "Create, edit and share videos at no cost in Google Vids." https://blog.google/products-and-platforms/products/workspace/google-vids-updates-lyria-veo/
7. Arxiv Cs Cy (2026-04-08). "EduIllustrate: Towards Scalable Automated Generation Of Multimodal Educational Content." https://arxiv.org/abs/2604.05005
8. Arxiv Cs Cy (2026-04-08). "A Multi-Agent Approach to Validate and Refine LLM-Generated Personalized Math Problems." https://arxiv.org/abs/2604.05160
9. EDUCAUSE Review (2026-04-08). "Building AI Initiatives with Bottom-Up Champions." https://er.educause.edu/articles/2026/4/building-ai-initiatives-with-bottom-up-champions
10. Atlantic (2026-04-10). "This Is What Fully Automated School Looks Like." https://www.theatlantic.com/technology/2026/04/ai-agents-school-education/686754/?utm_source=feed
11. Elearning Industry (2026-04-10). "AI-Generated Learning Content Governance: Ensuring Quality, Accuracy, And Trust." https://elearningindustry.com/ai-generated-learning-content-governance-ensuring-quality-accuracy-and-trust
12. Getting Smart (2026-04-13). "A Call to Action for AI to Promote Mathematical Reasoning." https://www.gettingsmart.com/2026/04/13/a-call-to-action-for-ai-to-promote-mathematical-reasoning/
13. Elearning Industry (2026-04-15). "The Case For Skills-Based Learning In A Rapidly Changing Workplace." https://elearningindustry.com/the-case-for-skills-based-learning-in-a-rapidly-changing-workplace
14. Elearning Industry (2026-04-21). "How AI Avatars Are Turning Static Documents Into Interactive Learning Experiences." https://elearningindustry.com/how-ai-avatars-are-turning-static-documents-into-interactive-learning-experiences
15. Arxiv Cs Cy (2026-04-21). "Bringing AI into the Classroom: A Structured Approach for Integrating AI into Software Engineering Education." https://arxiv.org/abs/2604.16328
16. Dev To Ai (2026-04-21). "Create Presentations from Any URL Automatically Using AI." https://dev.to/ai_presentation_d9935ff95/create-presentations-from-any-url-automatically-using-ai-5hho
17. Arxiv Cs Cy (2026-04-22). "Students Know AI Should Not Replace Thinking, but How Do They Regulate It? The TACO Framework for Human-AI Cognitive Partnership." https://arxiv.org/abs/2604.18737
18. Getting Smart (2026-04-22). "Responsible Inference Engines: Safeguarding Students with Learning Differences in the AI Era." https://www.gettingsmart.com/2026/04/22/responsible-inference-engines-safeguarding-students-with-learning-differences-in-the-ai-era/

19. EDUCAUSE Review (2026-04-23). "Teaching and Learning with AI: Three Priorities." https://er.educause.edu/podcasts/educause-exchange/teaching-and-learning-with-ai-three-priorities
20. The Register (2026-04-23). "Microsoft gives your Word documents an AI co-author you didn't ask for." https://go.theregister.com/feed/www.theregister.com/2026/04/23/microsoft_gives_your_word_documents/
21. Google DeepMind (2026-04-15). "Gemini 3.1 Flash TTS: the next generation of expressive AI speech." https://deepmind.google/blog/gemini-3-1-flash-tts-the-next-generation-of-expressive-ai-speech/
22. Arxiv Cs Cl (2026-04-24). "AFRILANGTUTOR: Advancing Language Tutoring and Culture Education in Low-Resource Languages with Large Language Models." https://arxiv.org/abs/2604.20996
23. Elearning Industry (2026-04-24). "AI Is Not Killing The Online Course Industry. It's Killing The Old Version Of It." https://elearningindustry.com/ai-is-not-killing-the-online-course-industry-its-killing-the-old-version-of-it
24. Wonkhe (2026-04-27). "AI literacy is not enough – universities must teach through disciplinary standards." https://wonkhe.com/blogs/ai-literacy-is-not-enough-universities-must-teach-through-disciplinary-standards/